# Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.

In partial fulfillment for the requirements in Capstone 1 and Research Subject


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# CHAPTER 1

# INTRODUCTION

## 1.1 Background of the Study

In today's highly competitive and technology-driven environment, industries are continuously seeking innovative ways to improve operational efficiency, reduce waste, and enhance productivity. The manufacturing sector, particularly in food production, plays a critical role in meeting consumer demand while maintaining quality and cost-effectiveness.- With the emergence of Industry 4.0, organizations are transitioning from traditional production methods to more advanced, automated, and data-driven systems.-

Industry 4.0 introduces the integration of digital technologies such as the real-time analytics, cloud computing,- and intelligent systems into manufacturing processes. These technologies enable organizations to collect, process, and analyze data instantly, allowing for improved decision-making and enhanced operational control. One of the key advantages of digital transformation is the ability to monitor production processes in real time, minimizing inefficiencies and preventing costly errors.

Despite these advancements, many small and medium-sized enterprises (SMEs), including AlDi Foods Inc., continue to rely on manual, paper-based methods of recording production data. These traditional systems are prone to human error, data inconsistencies, and delays in information processing. As a result, businesses experience data latency and lack real-time visibility into production performance, making it difficult to detect inefficiencies such as yield losses, machine issues, and material wastage at an early stage.

In food production, particularly in juice manufacturing, yield is a critical performance indicator. It represents the ratio of output (extracted juice) to input (raw materials). A high yield indicates efficient resource utilization, while a low yield suggests inefficiencies in the extraction process. Factors such as equipment condition, raw material quality, and operator performance significantly affect yield outcomes. Without a reliable monitoring system, these factors are difficult to analyze and optimize.

Currently, AIDi Foods Inc. relies on manual logging systems to track production data. This limits the company's ability to monitor extraction efficiency in real time. Production issues are often identified only after operations are completed, resulting in delayed decision-making and missed opportunities for immediate corrective actions.

To address these challenges, digitizing the production process is essential. By integrating automated data collection, and centralized software systems, production data can be monitored in real time. This enables organizations to transform raw data into actionable insights and supports proactive decision-making.

This study proposes the development of a **Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AIDi Foods Inc.** The system aims to replace manual recording methods with a centralized digital platform that provides real-time monitoring, analytics, and visualization of production data. Ultimately, the system seeks to improve efficiency, reduce waste, and enhance decision-making in food production operations.

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### 1.2 Statement of the Problem

The current production system of AlDi Foods Inc. relies heavily on manual, paper-based recording of production data. This approach presents several challenges that affect the efficiency, accuracy, and reliability of operations. Manual data recording consumes time, increases the possibility of human error, and limits the company's ability to monitor production performance in real time. As a result, production inefficiencies and extraction losses are often detected only after operations have been completed, making it difficult to implement immediate corrective actions.

Additionally, the absence of a centralized digital monitoring system severely limits management's access to timely and accurate production information. Without this visibility, leadership faces significant hurdles when making critical, time-sensitive decisions, evaluating overall employee productivity, and maintaining high operational efficiency.

Furthermore, this lack of real-time analytics hinders the company's capability to identify recurring production patterns and monitor extraction performance on the fly. As a result, the facility is left vulnerable to unaddressed inefficiencies and preventable material waste during its core manufacturing processes.

Specifically, this study seeks to answer the following questions:

1. How does manual recording of production data affect accuracy, consistency, and reliability at AlDi Foods Inc.?

2. What are the limitations of the current system in terms of real-time monitoring and data accessibility?
3. How can real-time monitoring improve the tracking of raw material inputs and production outputs?
4. How can yield losses during the extraction process be detected at an earlier stage?
5. How can inefficiencies in the production process be measured and analyzed effectively?
6. How can a digital system enhance managerial decision-making through timely and accurate data?
7. What features should be included in a digital production monitoring system to address existing challenges?

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## 1.3 Objectives of the Study

### 1.3.1 General Objective

The general objective of this study is to design and develop a digital production monitoring system for AlDi Foods Inc. that provides real-time yield analytics and extraction efficiency monitoring. This system specifically focuses on improving the overall accuracy, efficiency, and accessibility of production data.

To achieve this, the system aims to completely replace the company's current manual, paper-based recording processes. By implementing a centralized digital solution, the system will enable seamless, real-time tracking of production inputs and outputs, particularly within their juice extraction operations.

Specifically, this study seeks to address the critical need for operational visibility by providing a highly structured and automated framework for monitoring vital production indicators. In traditional manufacturing setups, tracking metrics such as raw material usage, total juice output, overall yield percentage, and extraction efficiency often relies on fragmented, manual processes. By integrating an automated tracking system, this research aims to centralize these disparate data points into a single, cohesive stream of real-time intelligence. This transition not only streamlines the collection of core metrics but also establishes a standardized baseline for performance evaluation, ensuring that every drop of resource and ounce of output is accounted for accurately.

Beyond mere data collection, the implementation of this automated system is strategically designed to mitigate the inherent risks of manual oversight. By significantly minimizing human error in data recording and eliminating the bureaucratic delays typical of legacy reporting methods, the system ensures that information moves at the speed of production. Consequently, this drastically improves the consistency, timeliness, and reliability of the data. Armed with high-fidelity, up-to-the-minute production insights, stakeholders and floor managers are empowered to make proactive, well-informed operational decisions that optimize throughput and reduce waste.

In addition, the system is designed to support real-time visualization and analysis of production performance through a centralized dashboard. This digital solution allows production managers to immediately identify variations in yield and extraction efficiency, detect possible losses during processing, and evaluate overall machine performance more effectively.

Ultimately, the study intends to enhance production monitoring capabilities and support data-driven decision-making within AlDi Foods Inc. By providing a practical and user-friendly interface, the system empowers leadership to transform raw operational data into actionable insights that optimize daily workflows.

### 1.3.2 Specific Objectives

Specifically, the study aims to:

1. Analyze the existing production process and identify inefficiencies in data recording and monitoring.
2. Design a user-friendly digital system that replaces manual production logs.
3. Develop a real-time monitoring feature for tracking raw material inputs and production outputs.
4. Implement automated yield calculation and analytics functionalities.
5. Monitor extraction efficiency and identify performance trends during production.
6. Provide an interactive dashboard for real-time visualization of production data.
7. Store and manage production data securely using a centralized database.
8. Evaluate the system's performance in terms of functionality, usability, accuracy, efficiency, and reliability.
9. Reduce delays in production monitoring by providing instant access to operational data and analytics.

10. Improve the company’s ability to identify extraction losses and production inefficiencies through automated monitoring and reporting features.
11. Support data-driven decision-making by generating accurate and timely production reports for management.
12. Enhance productivity and operational control through the integration of digital technologies and real-time monitoring tools.
13. Minimize manual recording errors and improve data consistency within production operations.
14. Provide a scalable foundation for future integration of IoT-based sensors and automated manufacturing technologies.

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**1.4 Scope and Delimitation**

This study focuses on the design and development of a digital production monitoring system for AlDi Foods Inc. The system is intended to improve the tracking, monitoring, and analysis of production data, particularly in terms of materials input, yield computation and extraction efficiency. The proposed system aims to replace manual recording methods with a centralized digital platform capable of providing real-time monitoring and analytical functionalities.

The scope of the study includes the development of features such as real-time production data input, automated yield calculation, extraction efficiency monitoring, dashboard visualization, and centralized database management. The system will allow authorized users to monitor production performance, track raw material usage, and evaluate production outputs through a digital interface. Additionally, the study includes basic analytics and visualization tools that support production analysis and operational decision-making.The system will be developed as a prototype intended for use within the operational context of AlDi Foods Inc. Data may be entered manually to demonstrate real-time monitoring capabilities. The study also covers system testing and evaluation in terms of functionality, usability, efficiency, accuracy, and reliability.

However, the study has several delimitations. The proposed system will not include full-scale industrial automation or advanced machine-learning capabilities. Integration with highly sophisticated Industrial Internet of Things (IIoT) infrastructures, robotics, or enterprise-level manufacturing systems is beyond the scope of this research. The study is limited only to monitoring production yield and extraction efficiency and does not cover other business operations such as inventory management, sales, accounting, payroll, or supply chain management.

Furthermore, the system will be tested in a controlled environment and may not fully represent all real-world production conditions. The study focuses only on the operational processes of AlDi Foods Inc., and the findings may not be fully generalized to other manufacturing companies with different production environments or operational requirements. Financial analysis, large-scale deployment, and long-term implementation assessment are also excluded from the scope of this study.

### 1.5 Significance of the Study

This study is beneficial to the following:

**AIDi Foods Inc.**

The system will improve production efficiency, reduce waste, and enhance overall productivity and profitability through real-time monitoring and analytics.

**Production Managers**

Managers will have access to timely and accurate production data, enabling faster decision-making, improved monitoring, and immediate response to operational issues.

**Employees**

The system will simplify production data recording processes, reduce manual workload, minimize human error, and improve operational efficiency.

**BSIT Students**

The study serves as a practical application of knowledge in system development, database management, software engineering, and real-time analytics.

**Future Researchers**

This study can serve as a reference for future research related to digital production systems, real-time monitoring, Industry 4.0 technologies, and manufacturing analytics.

**Manufacturing Industry**

The study may contribute to the advancement of digital transformation practices in manufacturing by demonstrating the benefits of real-time monitoring systems in improving efficiency and reducing production waste.

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## 1.6 Definition of Terms

**Analytics** – The process of analyzing data to extract meaningful insights and support decision-making.

**Centralized Database** – A digital storage system where production data is organized and managed in a single location for easier access and monitoring.

**Dashboard** – A graphical user interface that displays real-time production information, analytics, and system reports.

**Digitizing Production** – The conversion of manual production records and monitoring processes into digital format using information technology.

**Efficiency** – A measure of how effectively resources are utilized to produce desired outputs.

**Extraction** – The process of obtaining juice or liquid product from raw materials during production.

**Monitoring** – The continuous observation, tracking, and evaluation of production processes and operational performance.

**Real-Time** – Immediate processing, updating, and availability of data without significant delay.

**Yield** – The ratio of output produced compared to input materials, usually expressed as a percentage.

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# CHAPTER 2

## REVIEW OF RELATED LITERATURE AND STUDIES

### 2.1 Foreign Studies

[1]A study by Verhoef et al. (2021) analyzed the impact of transitioning from legacy paper-based logs to centralized digital dashboards in manufacturing. The researchers highlighted that manual data silos create "blind spots" or information gaps, where production anomalies—such as waste hotspots or mechanical inefficiencies—are often only discovered long after a batch is completed. By implementing a digitized analytics system, the facility achieved a more responsive production environment, allowing managers to make data-driven decisions based on live efficiency metrics rather than historical, error-prone logs. This study serves as a foundational reference for your claim that digitizing the production floor is a prerequisite for Industry 4.0 and is essential for reducing financial losses caused by avoidable process inefficiencies.

[2] A study by Makhmudah et al. (2025) explored the implementation of a digital production reporting system in a manufacturing environment. Using a qualitative case study approach, the researchers replaced manual reporting processes with a system integrated with sensors, automated data transmission, and dashboard visualization. The results showed improvements in data accuracy, operational efficiency, and decision-making. This study is relevant to the present research as it demonstrates the effectiveness of digital systems in improving production monitoring and data management.

-A study by Mansour and Al-Hamdani (2024) examined the application of internal Key Performance Indicators (KPIs) in evaluating production efficiency within the food industry, with a specific focus on non-conforming products identified during manufacturing operations. The study proposed a structured performance measurement system centered on the internal parts per million (INTppm) indicator, which quantifies defective units detected within production orders relative to the total number of items produced. This approach allows organizations to assess production quality at the operational level rather than relying solely on end-product inspection, thereby enabling more immediate and data-driven decision-making.

The authors emphasized that production performance in the food industry is highly dependent on continuous monitoring of quality-related indicators. In their framework, each production order is evaluated based on the number of non-conforming units identified during or after processing. The INTppm indicator serves as a quantitative measure of internal production defects, calculated by dividing the number of defective items by the total production volume for a given order and multiplying the result by one million. Through this method, the study provides a standardized way of expressing internal production inefficiencies, making it easier to compare performance across different production cycles, processes, or time periods.

In the case analysis presented in the study, data were collected from manufacturing operations involving food-related processing lines, where multiple production parameters were monitored. The researchers tracked the occurrence of non-conforming products across different operational activities and categorized them according to specific production conditions. These included manual and automated processes such as quality control checks, labeling procedures, measurement systems, and production line adjustments. The findings revealed that internal defect rates varied significantly depending on the type of operation being performed, indicating that certain stages of production were more prone to inefficiencies than others.

The study identified several operational issues that contributed to the increase in INTppm values. Among these were mechanical errors such as clamping inconsistencies, wear and tear of cutting tools, and misalignment during processing. These technical problems resulted in the production of defective units, which in turn affected overall production efficiency. In addition, the researchers observed that inadequate maintenance practices and delayed equipment replacement cycles further contributed to production instability. As a result, the internal defect rate became an important indicator not only of product quality but also of machine reliability and process stability.

To address these inefficiencies, Mansour and Al-Hamdani (2024) highlighted the importance of implementing corrective and preventive measures within the production system. These measures included improving machine reliability, enhancing process control systems, and implementing routine inspections at different stages of production. The study also recommended that operators conduct random quality checks during shifts to reduce the likelihood of defective outputs reaching later stages of production. Furthermore, the authors emphasized the importance of employee training programs, particularly for new workers, to ensure proper handling of equipment and adherence to standardized operating procedures.

Another key recommendation from the study was the development of structured training and monitoring systems that integrate skill evaluation checkpoints. These checkpoints allow management to assess worker competency and ensure that production tasks are performed consistently and correctly. The researchers also suggested strengthening workforce coordination and communication to improve overall operational stability. By stabilizing human resource performance alongside technical improvements, the production system becomes more resilient to errors and variability.

The study also discussed the role of internal KPIs in identifying patterns of inefficiency over time. By continuously monitoring INTppm values across production orders, organizations can detect recurring problems and implement targeted interventions. This long-term data collection approach enables companies to shift from reactive problem-solving to proactive process optimization. Instead of addressing defects only after they occur, managers can anticipate potential issues based on historical KPI trends and adjust production parameters accordingly.

In addition to operational improvements, the authors emphasized that internal KPIs such as INTppm are essential tools for strategic decision-making in manufacturing environments. These indicators provide a clear link between production performance and business outcomes, such as cost efficiency, waste reduction, and product quality. High internal defect rates not only increase production costs but also reduce overall system efficiency, making KPI-based monitoring an essential component of modern manufacturing management systems.

The findings further demonstrated that although internal defect rates (INTppm) were relatively high in some production stages, they could be reduced through systematic process improvements. The study showed that many production inefficiencies were not due to external factors but were instead related to controllable internal variables such as equipment condition, operator behavior, and process design. This highlights the importance of internal performance measurement systems in identifying root causes of inefficiencies within production environments.

Overall, Mansour and Al-Hamdani (2024) concluded that the use of internal KPIs such as INTppm provides a reliable and effective method for evaluating production efficiency in the food industry. The study supports the idea that manufacturing performance should be assessed not only based on final output quality but also through continuous monitoring of internal process deviations. By integrating KPI systems into production management, organizations can improve operational efficiency, reduce waste, and enhance overall product quality.

This literature is relevant to the present study as it provides a strong foundation for the use of internal performance indicators, particularly INTppm, in identifying production inefficiencies and improving quality control systems. It also supports the development of data-driven monitoring frameworks for food manufacturing processes, where real-time tracking of defects and process performance is essential for achieving operational excellence.-

-A comprehensive review conducted by Aftab Siddique, Ashish Gupta, Jason T. Sawyer, Tung-Shi Huang, and Amit Morey (2025), entitled *Big Data Analytics in Food Industry: A State-of-the-Art Literature Review*, examined the increasing influence of artificial intelligence (AI), machine learning (ML), and big data analytics on the modernization of food industry operations. Published in *npj Science of Food*, the study explored numerous scholarly works related to digital transformation and data-driven technologies in food production, processing, quality assurance, supply chain management, and food safety. The researchers emphasized that the food industry is undergoing a significant technological shift as organizations increasingly adopt digital systems capable of collecting, storing, processing, and analyzing large volumes of operational data. As production processes become more complex and consumer demands continue to evolve, the use of advanced analytics has become an important strategy for improving efficiency, productivity, and competitiveness.

According to the authors, modern food manufacturing facilities generate substantial amounts of data from various stages of production, including raw material handling, processing, packaging, storage, distribution, and quality control. Traditionally, much of this data remained underutilized due to limitations in data collection and analysis methods. However, advancements in digital technologies have enabled organizations to transform raw production data into valuable information that can support decision-making and operational improvements. The study explained that big data analytics provides organizations with the capability to identify patterns, detect inefficiencies, forecast outcomes, and monitor performance more accurately than conventional approaches.

The review highlighted that artificial intelligence and machine learning technologies have become essential tools for enhancing food production systems. These technologies allow organizations to process large datasets and generate insights that would be difficult to obtain through manual analysis. AI-based models can learn from historical and real-time data, making it possible to predict trends, identify operational bottlenecks, and optimize production processes. Through continuous analysis of production information, organizations can improve process control, reduce waste, and enhance overall manufacturing efficiency.

One of the major areas discussed in the study was the application of predictive analytics in food processing operations. Predictive analytics utilizes historical production data and advanced algorithms to forecast future outcomes and support proactive decision-making. The authors noted that predictive models can be applied to numerous food processing activities, including drying, baking, fermentation, extrusion, freezing, and product quality assessment. By analyzing operational data, these systems can predict product characteristics, estimate processing times, and identify optimal operating conditions. Such capabilities help manufacturers improve consistency, reduce production errors, and maximize resource utilization.

The researchers also emphasized the importance of real-time monitoring systems in modern food manufacturing environments. Traditional production monitoring methods often rely on manual recording procedures that can be time-consuming and susceptible to human error. In contrast, digital monitoring systems continuously collect and analyze operational data, allowing managers and operators to observe production performance as activities occur. Real-time monitoring provides immediate visibility into production processes and enables organizations to respond quickly to operational issues. The study suggested that organizations adopting real-time analytics can achieve greater efficiency because production deviations and performance problems can be detected and corrected before they significantly affect output quality or productivity.

Another significant finding of the review was the growing use of artificial neural networks in food processing applications. Artificial neural networks are machine learning models designed to mimic certain functions of the human brain by learning patterns from data. The study reported that neural networks have been successfully applied in predicting moisture content, temperature variations, shelf life, drying kinetics, fermentation performance, product texture, and other critical food processing parameters. These models can analyze complex relationships among multiple variables and generate highly accurate predictions, making them valuable tools for process optimization and quality management.

The authors further discussed the integration of computer vision systems and image-processing technologies in food manufacturing. Computer vision enables machines to analyze visual information and automatically inspect products during production. Various studies reviewed by the authors demonstrated how image analysis systems can be used for food grading, defect detection, spoilage identification, quality evaluation, and product classification. These technologies reduce the need for manual inspection while increasing consistency and accuracy in quality assessment procedures. The implementation of computer vision technologies contributes to improved productivity and enhanced product quality by enabling faster and more reliable inspection processes.

In addition to production monitoring and quality control, the review explored the role of big data analytics in food safety management. Food safety remains one of the most critical concerns in the food industry, and the study highlighted several ways in which AI and machine learning can strengthen safety practices. Advanced analytical systems can identify contamination risks, detect spoilage indicators, analyze consumer feedback, and monitor disease outbreaks associated with food products. By utilizing large datasets from multiple sources, organizations can improve traceability and respond more effectively to food safety concerns. These capabilities contribute to the protection of consumers while also reducing economic losses associated with product recalls and contamination incidents.

The researchers also examined the use of machine learning algorithms for demand forecasting and supply chain optimization. Accurate forecasting is essential for maintaining production efficiency and minimizing waste. The study noted that AI-powered forecasting models can analyze historical sales data, market trends, seasonal patterns, and consumer behavior to predict future demand more accurately. Improved forecasting enables manufacturers to optimize inventory levels, schedule production activities efficiently, and allocate resources more effectively. As a result, organizations can reduce operational costs while maintaining adequate product availability.

Another important aspect highlighted in the review was the contribution of AI technologies to sustainability and resource management. The food industry faces increasing pressure to reduce waste, improve energy efficiency, and minimize environmental impacts. According to the authors, data-driven systems can assist organizations in achieving sustainability objectives by identifying inefficiencies and recommending process improvements. AI-based analytics can optimize the use of raw materials, energy, water, and labor resources, thereby reducing operational waste and supporting environmentally responsible production practices. These benefits align with global efforts to promote sustainable manufacturing and resource conservation.

The study also recognized several challenges associated with the implementation of big data analytics in food manufacturing environments. Although digital technologies offer significant advantages, organizations may encounter obstacles related to data quality, system integration, infrastructure requirements, cybersecurity, and workforce readiness. Effective implementation requires reliable data collection mechanisms, adequate technological resources, and personnel capable of interpreting analytical outputs. The authors emphasized that organizations must develop appropriate strategies to address these challenges and maximize the benefits of digital transformation initiatives.

Furthermore, the review concluded that the future of the food industry will increasingly depend on intelligent systems capable of transforming production data into actionable knowledge. As artificial intelligence, machine learning, and big data technologies continue to evolve, food manufacturers are expected to adopt more advanced analytical tools that support automated decision-making, predictive maintenance, process optimization, and continuous performance improvement. The integration of these technologies is anticipated to enhance operational efficiency while enabling organizations to adapt more effectively to changing market conditions and consumer demands.

This study is highly relevant to the present capstone project entitled *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* because it provides strong theoretical and practical support for the implementation of digital monitoring and analytics systems within food manufacturing operations. The review demonstrates how real-time data collection, predictive analytics, and artificial intelligence can be utilized to improve production visibility, evaluate operational performance, and support informed decision-making. Similar to the proposed capstone project, the studies reviewed by Siddique et al. (2025) emphasize the importance of transforming production data into meaningful information that can be used to enhance efficiency and productivity.

The findings particularly support the development of a system capable of monitoring production yield and extraction efficiency in real time. Through continuous data collection and analysis, management can identify process variations, evaluate extraction performance, detect production losses, and implement corrective actions promptly. The concepts presented in the review validate the significance of digital transformation initiatives within food processing organizations and reinforce the value of data-driven decision-making in improving operational outcomes. Therefore, the study serves as a strong foundation for the proposed system by demonstrating how advanced analytics technologies can contribute to more efficient, accurate, and effective production management within the food industry.

-A study by Rivadeneira et al. (2023), published in *Food Research*, examined the production of pectin from “saba” banana peel waste using ultrasound-assisted extraction and evaluated its performance as an emulsifying agent compared with commercial low-methoxyl pectin (LMP). The study emphasized the potential of agricultural waste valorization by converting banana peels into functional food ingredients suitable for emulsion-based systems. Findings revealed that the extracted pectin had lower equivalent weight and a reduced degree of esterification than commercial LMP, indicating differences in molecular structure that influence its functional behavior in food applications. Although UEP showed lower purity, it contained relatively higher amounts of protein and ash, which contributed positively to its interfacial and emulsifying properties.

Structural and physical analyses further confirmed these differences, where FTIR results indicated variations in esterified carboxyl groups between UEP and LMP. Microscopic observation showed that emulsions stabilized by UEP formed smaller and more evenly distributed oil droplets compared to those stabilized by commercial pectin. This droplet size reduction contributed to improved emulsion stability, as it enhanced interfacial coverage and minimized phase separation. The study also reported that changes in formulation factors such as pH level, oil concentration, and ionic strength significantly affected emulsion behavior, with UEP consistently showing stronger adaptability under varying conditions.

In terms of functional performance, emulsions containing UEP exhibited higher viscosity and better resistance to breakdown compared to those stabilized by LMP. The study also found that UEP influenced lipid digestion, as it reduced free fatty acid release during in vitro digestion, suggesting a potential role in controlling lipid bioaccessibility. Overall, the researchers concluded that ultrasound-extracted pectin from banana peel waste is a promising sustainable alternative to commercial pectin, offering improved emulsifying stability and functional benefits for food formulation and processing systems.-

**2.2 Local Studies**

[3]A study by Bagaforo and Albason (2024), presented at the 19th European Conference on Innovation and Entrepreneurship (ECIE), investigated the digital transformation journeys of Micro, Small, and Medium Enterprises (MSMEs) in Iloilo Province, identifying the specific enablers and barriers local businesses face when adopting new technologies. The research utilized a descriptive-correlational approach to evaluate how local infrastructure and workforce "readiness" influence the transition from traditional to digitized operations. Their findings highlighted that while digital tools significantly enhance local business competitiveness and operational efficiency, success is often hindered by technical literacy gaps and varying levels of infrastructural support. This study provides critical contextual evidence for AIDi Foods Inc. regarding the feasibility of implementing automated systems within the local economic landscape, directly supporting the Significance of the Study by illustrating how digital integration optimizes resource management and regional market positioning.

[4]A study by De Leon et al. (2017), archived in the Central Philippine University (CPU) Institutional Repository, focused on the design and development of an online ordering system

for Iloilo Supermart using the V-Model Software Development Life Cycle (SDLC). The researchers implemented a secure, centralized database and a user-friendly management dashboard to streamline operations for the Iloilo-based food retailer, effectively mitigating risks associated with physical data damage and manual record-keeping.

-A study conducted by Palero et al. (2026), titled *“Green Extraction and Sustainability Assessment of Total Carotenoids from Carrot (Daucus carota) Peel Waste using Soybean Oil,”* explored a sustainable and environmentally responsible approach for recovering valuable bioactive compounds from agricultural waste materials. Published in the Davao Research Journal, the study primarily addressed the growing concern over the environmental and health hazards associated with conventional solvent-based extraction methods, particularly those involving organic solvents such as hexane. These traditional solvents, while efficient in extracting carotenoids, are widely criticized due to their toxicity, flammability, and environmental persistence. In response to these limitations, the researchers investigated soybean oil as a safer, biodegradable, and food-grade alternative solvent capable of efficiently extracting carotenoids from carrot peel waste while aligning with green chemistry and circular economy principles.

Carotenoids are naturally occurring pigments responsible for the orange, yellow, and red coloration in many fruits and vegetables, and they are widely recognized for their antioxidant, anti-inflammatory, and health-promoting properties. Because of their high commercial value and wide applicability in food, pharmaceutical, and cosmetic industries, there is increasing global interest in improving their extraction from natural sources, especially from agricultural by-products that are often discarded. Carrot peels, which are generated in large quantities during food processing, represent a significant yet underutilized source of carotenoids. The study emphasized that valorizing such waste materials not only contributes to resource efficiency but also reduces environmental pollution and supports sustainable production systems.

To achieve its objectives, the researchers employed a two-factor, three-level full factorial experimental design to systematically evaluate and optimize the extraction process. The independent variables considered were the liquid-to-solid (L/S) ratio and extraction temperature, both of which are known to significantly influence mass transfer and solute diffusion during extraction processes. The dependent variable was the total carotenoid content (TCC), expressed in micrograms of β-carotene per gram of dried carrot peel. The experimental procedure involved freeze-drying carrot peels to preserve bioactive compounds, followed by pulverization to increase surface area and enhance solvent penetration. The prepared samples were then mixed with soybean oil under controlled conditions and subjected to agitation using a shaking water bath. This setup ensured consistent contact between the solvent and plant material, facilitating efficient extraction of carotenoids into the oil phase.

Quantitative analysis of carotenoid content was performed using UV-Visible spectrophotometry at 450 nm, applying the Beer-Lambert law for concentration determination. The study also implemented rigorous statistical tools to evaluate the reliability and validity of the experimental model. These included analysis of variance (ANOVA), regression modeling, and diagnostic tests such as the Shapiro-Wilk test for normality, Durbin-Watson test for autocorrelation, Cook's distance for identifying influential outliers, and Variance Inflation Factor (VIF) for detecting multicollinearity. Additionally, model performance was assessed using $R^2$, adjusted $R^2$, and predicted $R^2$ values to ensure both accuracy and predictive capability.

The experimental results demonstrated that both the liquid-to-solid ratio and temperature significantly influenced carotenoid extraction efficiency, although their effects were not equal in magnitude. The highest observed total carotenoid content was 84.22 ± 5.56 μg β-carotene/g of dried peel, achieved at a liquid-to-solid ratio of 40 mL/g and a temperature of 45 °C. This result indicated that increasing the solvent proportion relative to the biomass significantly enhances extraction efficiency by improving concentration gradients and promoting better diffusion of carotenoids from the plant matrix into the oil solvent. In contrast, temperature exhibited a positive but comparatively weaker effect on extraction performance, suggesting that thermal enhancement plays a supporting role rather than a dominant one in the process.

Statistical analysis confirmed that the overall model was highly significant ($p < 0.0001$), with a strong coefficient of determination ($R^2 = 0.95$), indicating that the model effectively explained most of the variability in carotenoid recovery. The adjusted $R^2$ (0.94) and predicted $R^2$ (0.93) values further supported the robustness and predictive reliability of the model. However, a statistically significant lack of fit was also observed, suggesting that certain complex interactions or nonlinear behaviors within the system were not fully captured by the model. This indicates that while the model is strong in predictive capability, there remains room for refinement to improve its descriptive accuracy.

Through response surface methodology optimization, the study identified the most favorable extraction conditions as a liquid-to-solid ratio of 40 mL/g and a temperature of 30 °C. Under these conditions, the predicted carotenoid yield was 77.49 ± 13.13 μg β-carotene/g of dried peel. To validate the model, experimental runs were conducted under the same optimized conditions, yielding an actual average value of 76.04 ± 3.71 μg β-carotene/g. The small deviation between predicted and experimental values, reflected in a low relative error, confirmed the reliability and practical applicability of the developed model. This close agreement demonstrates that the optimization approach used in the study is effective for predicting carotenoid extraction performance under varying process conditions.

Further analysis revealed that the liquid-to-solid ratio was the most influential factor affecting carotenoid recovery. This can be attributed to its direct effect on mass transfer dynamics. A higher solvent volume increases the concentration gradient between the solid matrix and the liquid phase, thereby enhancing diffusion rates and preventing early saturation of the solvent. This allows more carotenoids to be released from the plant material into the soybean oil. While temperature also contributes by improving solubility and weakening cellular structures, its effect becomes limited once the system approaches equilibrium or when saturation conditions are reached. In oil-based extraction systems, this suggests that solvent quantity plays a more critical role than thermal energy in determining overall extraction efficiency.

The study also compared its findings with previous research involving other biomass sources such as pumpkin, peach, and carrot byproducts. These comparisons showed consistent trends, where higher solvent-to-solid ratios generally resulted in improved carotenoid recovery.

However, excessive temperature increases were often associated with reduced efficiency due to possible degradation of heat-sensitive carotenoid compounds. This reinforces the idea that moderate temperature conditions combined with optimized solvent ratios are ideal for maximizing yield in green extraction systems.

In addition to extraction efficiency, the study conducted a sustainability evaluation using the Path2Green assessment framework, which measures the environmental performance of extraction processes across 12 standardized principles. These principles include biomass sourcing, transport, pretreatment, solvent selection, scaling, purification, energy use, and waste management. The overall Path2Green score obtained was 0.593 out of a possible maximum of 1.00, indicating a moderately sustainable and environmentally friendly process.

The sustainability assessment revealed that the process performed strongly in several key areas. High scores were recorded for biomass utilization, as the study effectively repurposed agricultural waste rather than relying on freshly cultivated crops. Similarly, solvent selection scored highly due to the use of soybean oil, which is biodegradable, non-toxic, non-volatile, and safe for food-related applications. The process also demonstrated strong performance in scalability, purification, post-treatment, application potential, and waste management, as the extracted product could be directly utilized without requiring extensive purification or solvent removal.

However, certain limitations were identified in the sustainability analysis. Energy consumption received a low score due to the use of freeze-drying, grinding, and controlled temperature water bath systems, all of which require significant energy input. Pretreatment processes also contributed to reduced sustainability performance because mechanical processing steps were necessary to prepare the biomass for extraction. Additionally, yield efficiency was identified as an area for improvement, indicating that the extraction process was not fully exhaustive and that some carotenoids likely remained unextracted in the residual biomass. The study also noted the absence of a closed-loop extraction system, which limited its circular economy potential.

Despite these limitations, the study highlighted that soybean oil-based extraction remains a promising green alternative to conventional solvent systems. The integration of environmentally friendly solvents with waste valorization strategies demonstrates a strong alignment with sustainable development goals and green chemistry principles. Furthermore, the extracted carotenoid-rich oil can potentially be applied directly in food formulations, cosmetics, pharmaceuticals, and nutraceutical products without requiring further purification, thereby reducing processing costs and environmental burden.

In conclusion, Palero et al. (2026) successfully demonstrated that soybean oil is an effective and sustainable solvent for carotenoid extraction from carrot peel waste. The optimized extraction conditions provided high yields with strong model reliability, while the sustainability assessment confirmed its moderate environmental performance. Although improvements are still needed in energy efficiency, extraction completeness, and system integration, the study provides strong evidence that agricultural waste can be effectively transformed into high-value bioactive compounds using green extraction technologies. This research contributes significantly to the growing field of sustainable bioprocessing and supports the development of eco-friendly industrial practices that prioritize both efficiency and environmental responsibility.

-A study by **Manalo (2021)** discussed the role of Digital Agriculture in supporting the modernization and industrialization goals of the Philippine agricultural sector through the adoption of Information and Communications Technologies (ICTs). The study highlighted how digital technologies have become important tools in improving agricultural productivity, efficiency, and access to information. Various ICT-based innovations have already been utilized in agriculture, including web-based disease diagnosis systems, crop monitoring sensors, mobile applications, and digital trading platforms. In the Philippines, DA-PhilRice developed several ICT-driven applications such as the Leaf Color Chart (LCC) app, e-binhi, e-damuhan, and AgriDoc, which provide farmers with timely and relevant agricultural information. These technologies were designed to support better crop management, improve decision-making, and enhance overall agricultural operations. The study emphasized that digital technologies can help farmers access critical information related to crop production, nutrient management, pest control, and market opportunities, enabling them to make informed decisions and improve farm performance. Through these innovations, Digital Agriculture has been recognized as a significant strategy in transforming traditional farming practices into more efficient, data-driven, and technology-enabled systems.

The study further examined the level of ICT adoption among Filipino farmers and identified several challenges that hinder the effective use of digital technologies. Based on the results of the 2016–2017 Rice-Based Farm Households Survey, approximately 93 percent of rice-farming households had access to mobile phones, smartphones, and internet services. Similarly, a large percentage of farmers expressed willingness to receive agricultural information through text messaging and internet-based platforms. Despite this high level of access, actual utilization of ICTs for farming activities remained relatively low. The survey revealed that only 31 percent of farmers used text messaging as a source of agricultural information, while only 21 percent actively searched for farming-related information online. These findings suggest that access to technology alone does not guarantee its effective use. The study argued that several factors contribute to this gap, including low levels of digital literacy, limited exposure to technology, lack of confidence in using digital tools, and ICT anxiety among farmers.

According to Manalo (2021), ICT anxiety is one of the major barriers to technology adoption in agriculture. ICT anxiety refers to the discomfort, fear, or uncertainty experienced by individuals when interacting with computers, mobile devices, or other digital technologies. This challenge is particularly common among older populations, including many Filipino farmers. Since a significant portion of the agricultural workforce in the Philippines consists of aging farmers, the adoption of new technologies often becomes difficult. The study explained that limited experience with digital tools and insufficient technical knowledge can discourage farmers from utilizing ICT-based solutions, even when such technologies are readily available. As a result, many farmers continue to rely on traditional methods of obtaining information, such as consulting fellow farmers, agricultural extension workers, and local community networks. This preference for interpersonal communication demonstrates that technological innovations must be accompanied by strategies that address behavioral and educational barriers to ensure successful implementation.

To address these challenges, the study introduced the concept of infomediaries as a means of bridging the gap between technology and farmers. Infomediaries are individuals who facilitate access to information by helping others use digital tools and communication technologies. The study described infomediaries as important agents in reducing the digital divide, particularly among populations that have limited technological skills. The digital divide refers to the disparity between individuals who can effectively access and utilize digital technologies and those who cannot. This divide is influenced by various factors, including educational attainment, socioeconomic status, technological literacy, and access to resources. By mobilizing infomediaries, agricultural organizations can help ensure that the benefits of digital technologies reach a wider population of farmers, including those who may be hesitant or unable to use ICTs independently.

The study highlighted the implementation of the Infomediary Campaign by DA-PhilRice from 2012 to 2017 as an innovative approach to promoting digital inclusion in agriculture. The campaign engaged high school students from rice-farming communities and encouraged them to serve as infomediaries for their parents and local farmers. These students accessed agricultural information through digital platforms and shared the information with farmers who lacked the skills or confidence to use ICT tools themselves. Through this initiative, young people became active participants in agricultural development while helping bridge the information gap within their communities. The campaign demonstrated how youth engagement can contribute to the dissemination of agricultural knowledge and encourage greater acceptance of digital technologies among farmers.

Results of the Infomediary Campaign showed positive outcomes in promoting information access and technology adoption. The study reported that more than 4,000 student-infomediaries submitted nearly 20,000 inquiries to the PhilRice Text Center between 2012 and 2016. These inquiries covered a wide range of agricultural topics, including rice varieties, pest management, nutrient management, crop establishment, and general farming practices. Students not only gathered information but also actively shared their knowledge with their parents, relatives, and other members of the farming community. The study observed that information dissemination often extended beyond the immediate recipients, creating a multiplier effect where agricultural knowledge was continuously shared among farmers. This process contributed to improved awareness of modern farming techniques and increased utilization of available information resources.

Furthermore, the study found evidence that the campaign positively influenced farmers' information-seeking behavior. Farmers who were exposed to the initiative became more likely to use text messaging services and digital communication platforms to obtain agricultural information. In some communities, the habit of consulting digital information sources continued even after the campaign ended. These findings suggest that targeted interventions and support systems can help overcome barriers to technology adoption and encourage the long-term use of ICT-based agricultural solutions. The study emphasized that successful digital transformation requires not only technological infrastructure but also educational programs, capacity-building initiatives, and community support mechanisms that enable users to maximize the benefits of digital tools.

The study concluded that Digital Agriculture can only achieve its full potential if efforts are made to ensure inclusivity and accessibility among all stakeholders. While digital technologies offer significant opportunities for improving productivity, information management, and decision-making, their effectiveness depends largely on the willingness and ability of users to adopt them. Initiatives such as the Infomediary Campaign demonstrate that human-centered approaches can play a critical role in facilitating technology adoption and reducing barriers associated with digital literacy and ICT anxiety. The study recommended expanding infomediary programs, strengthening ICT training initiatives, integrating agriculture-related digital skills into educational curricula, and providing continuous support to both farmers and agricultural extension workers.

This literature is relevant to the present study as it demonstrates the significant role of digital technologies in improving operational efficiency, information management, and data-driven decision-making. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights how digital solutions can enhance monitoring processes, facilitate timely access to information, and support more effective management practices. The findings support the development of digital production systems by emphasizing the value of technology in collecting, analyzing, and utilizing data to improve productivity and operational performance. Furthermore, the study reinforces the importance of user adoption and accessibility in ensuring the successful implementation of digital innovations within agricultural and food-processing industries.- left

-A study by **Barroga, Rola, Depositario, Digal, and Pabuayon (2019**) investigated the determinants of the extent of technological innovation adoption among micro, small, and medium food processing enterprises (MSMFEs) in the Davao Region, Philippines. The research focused on enterprises that were beneficiaries of the Department of Science and Technology–Small Enterprises Technology Upgrading Program (DOST-SETUP), a government initiative that provides technological support, upgraded equipment, and technical assistance to improve productivity and competitiveness in the food processing sector. The study aimed to determine not only whether MSMFEs adopt technological innovations but also the degree or extent to which these innovations are implemented within their operations.

The researchers utilized a census of 52 MSMFEs in Davao Region and employed both quantitative and qualitative methods to analyze the data. Specifically, an ordered logistic regression model was used to identify the factors influencing the level of technological innovation adoption. The study also used index construction to measure the extent of adoption across multiple technological interventions, including food safety training, Good Manufacturing Practices (GMP) assessment, plant layout improvement, packaging and labeling, laboratory testing, product development, energy audit, cleaner production technology (CPT) consultancy, MPEX consultancy, and technology training. These innovations represent a comprehensive package of technical and managerial upgrades introduced through the SETUP program.

Findings of the study showed that MSMFEs in the Davao Region generally exhibited varying levels of technological innovation adoption, ranging from non-adoption to full adoption. A small portion of the enterprises failed to adopt any of the technological innovations provided, while others demonstrated low to moderate adoption levels. However, the majority of MSMFEs were classified as highly adopters, indicating that most enterprises were able to integrate several SETUP innovations into their operations. Only a smaller segment of the sample achieved full adoption, meaning they implemented nearly all of the technological innovations offered by the program.

Among the specific innovations introduced by SETUP, food safety training emerged as the most widely adopted intervention, followed by GMP assessment and MPEX consultancy. These innovations were particularly important for MSMFEs because they directly address compliance with food safety standards and operational efficiency. On the other hand, innovations such as product development and cleaner production technology consultancy were among the least adopted. The study noted that more advanced or resource-demanding innovations tended to have lower adoption rates, especially among smaller enterprises with limited capacity and technical expertise.

The study further highlighted that the adoption of technological innovations does not only depend on whether enterprises accept or reject a particular technology, but also on the depth and breadth of implementation across various business functions. This means that even if MSMFEs adopt certain innovations, the level of integration into their production processes, management systems, and product development activities may differ significantly. Based on this, the researchers categorized MSMFEs into five groups according to their extent of adoption: did not adopt, less adopted, moderately adopted, highly adopted, and fully adopted.

In terms of distribution, most medium-sized enterprises were found to have lower levels of adoption compared to micro and small enterprises. Micro enterprises, despite their limited resources, showed a relatively higher tendency toward adopting multiple technological innovations. Small enterprises were mostly classified as moderate to high adopters, depending on their exposure to export markets and access to external support services. The study suggests that enterprise size alone does not fully determine innovation adoption, but interacts with other organizational and contextual factors.

One of the key findings of the study is that the combination of food safety training, GMP assessment, and MPEX consultancy was the most commonly adopted set of innovations. These three are closely interconnected, as MPEX consultancy often recommends improvements related to GMP compliance and food safety practices. This indicates that MSMFEs tend to adopt innovations in clusters rather than individually, especially when the technologies are complementary and collectively improve production efficiency and compliance with regulatory standards.

The study also revealed that full adoption of technological innovations was relatively limited, with only about 21% of MSMFEs achieving this level. Enterprises under this category were able to implement most or all of the SETUP-supported innovations, demonstrating strong organizational capacity and readiness for technological upgrading. Micro enterprises in this category adopted nearly all available innovations, while small enterprises typically adopted a slightly smaller set, often excluding more advanced innovations such as laboratory analysis or product development.

Highly adopted MSMFEs represented a significant portion of the sample. These enterprises generally implemented key foundational innovations such as GMP assessment and food safety training, along with selected productivity-enhancing interventions. However, they still lacked full integration of more advanced technological services. This suggests that while these enterprises are relatively progressive in adopting innovation, constraints such as cost, technical expertise, and organizational capacity still limit full adoption.

Moderately adopted MSMFEs showed partial integration of SETUP innovations, often focusing on basic operational improvements. These enterprises typically adopted essential interventions such as food safety training and plant layout improvements but did not fully engage in more complex innovations like laboratory analysis or product development. Their level of adoption indicates an intermediate stage of technological readiness, where awareness and initial implementation exist but full assimilation remains limited.

Less-adopted MSMFEs were characterized by minimal engagement with SETUP innovations, typically adopting only a few interventions such as technology training or food safety training. In this category, medium enterprises were more dominant, suggesting that some established firms may perceive less need for external technological assistance due to existing internal capabilities. This group highlights that adoption is not always higher among larger firms, especially when they already possess in-house technical expertise.

At the lowest level, the study identified a very small number of enterprises that did not adopt any technological innovations at all. These cases were linked to external shocks such as natural disasters and management issues, which disrupted operations and prevented the proper utilization of SETUP assistance. These findings emphasize that external environmental factors and governance issues can also significantly affect innovation adoption outcomes.

The study further examined the determinants of technological innovation adoption using an ordered logistic regression model. The results identified four statistically significant factors influencing the extent of adoption: sex of the owner, educational attainment, business scale, and type of market. These variables were found to consistently influence whether MSMFEs adopt innovations at a higher or lower level.

In terms of sex of the owner, the study found that male-owned enterprises were more likely to achieve higher levels of technological innovation adoption compared to female-owned enterprises. Although female entrepreneurs represented a larger proportion of MSMFE owners in the sample, male owners were more likely to engage in extensive adoption. This was attributed to differences in mobility, access to training opportunities, and exposure to external information networks, which may influence innovation-related decision-making.

Educational attainment was also found to have a positive relationship with the extent of adoption. MSMFE owners with higher educational levels were more likely to adopt a wider range of technological innovations. This is because education enhances the ability to understand technical information, evaluate innovation benefits, and manage the risks associated with technological change. As a result, more educated owners tend to be more open to upgrading their production systems and integrating new technologies into their operations.

Business scale, on the other hand, showed a negative relationship with adoption extent, indicating that micro enterprises were more likely to adopt innovations compared to small and medium enterprises. This finding suggests that smaller firms are more flexible and can implement changes more quickly due to simpler organizational structures and centralized decision-making. In contrast, larger firms may experience more complex internal processes that slow down innovation adoption.

The type of market was also identified as a significant determinant. MSMFEs that cater to both domestic and export markets were more likely to adopt technological innovations at higher levels. This is because export-oriented firms are required to comply with stricter international standards, which encourages them to adopt advanced technologies and quality control systems to remain competitive in global markets.

Other variables such as age, years of experience, organizational type, and sources of information were found to be statistically insignificant in influencing the extent of technological innovation adoption. This suggests that while these factors may have some descriptive influence, they do not significantly determine the level of adoption when analyzed together with stronger predictors such as education, gender, business scale, and market orientation.

Overall, the study concluded that technological innovation adoption among MSMFEs in the Davao Region is a multi-dimensional process influenced by both individual and organizational factors. The findings highlight that innovation adoption is not simply a matter of access to technology, but also depends on the characteristics of decision-makers and the market environment in which enterprises operate.

The study is relevant to the present research because it provides empirical evidence on how personal demographics and organizational characteristics influence the extent of technological innovation adoption in food processing enterprises. It also highlights the importance of government support programs such as DOST-SETUP in promoting innovation among MSMEs, particularly in developing regions.

A study by **Ang, Amongo, Paras Jr., Suministrado, and Peralta (2022),** entitled *Development of Banana Peduncle Juice Extractor for Ethanol and Fiber Production*, focused on the conceptualization, design, fabrication, and performance evaluation of a specialized mechanical system intended for the efficient processing of banana (Musa acuminata C.) peduncles. The research was motivated by the increasing concern regarding agricultural waste management in the Philippines, particularly within large-scale banana plantations where peduncles are typically discarded despite their significant potential as raw materials for bioethanol production and natural fiber utilization. The authors emphasized that although banana peduncles contain valuable lignocellulosic components and fermentable sugars, their utilization remains limited due to the lack of appropriate extraction technology that can simultaneously maximize juice recovery while preserving fiber integrity. Conventional extraction machines, originally designed for other agricultural commodities such as sugarcane, were found to be inefficient when applied to banana peduncles due to differences in material structure, density, and mechanical resistance. This gap in technology prompted the development of a dedicated juice extraction system tailored specifically for banana peduncle processing.

The developed machine was designed using engineering principles that prioritized functionality, manufacturability, safety, and cost-effectiveness. The researchers deliberately utilized locally available materials in order to ensure affordability and ease of replication, particularly for potential use in rural agricultural communities and small-scale processing facilities. The system employed a crushing roll mechanism as the primary extraction component, consisting of a main roll and multiple feeder rolls arranged in a configuration that enables controlled compression of the peduncle material. This configuration was intended to optimize juice release while minimizing excessive mechanical damage to the fiber structure, thereby allowing the biomass to be utilized for both ethanol fermentation and fiber recovery applications. The structural design also incorporated protective enclosures and adjustable components to enhance operational safety and accommodate varying operator conditions. In addition, the transmission system was engineered using a combination of engine-driven pulleys, gear reducers, and chain-sprocket assemblies, allowing controlled variation of rotational speed to evaluate performance under different operational settings.

To assess the effectiveness of the developed machine, the researchers employed a systematic experimental design using a two-factor, three-level full factorial approach. The two independent variables considered were roll rotational speed (50 rpm, 70 rpm, and 90 rpm) and crushing load (0.8 kN, 1.2 kN, and 1.6 kN). A total of 27 experimental runs were conducted to determine the influence of these variables on machine performance. Each trial utilized approximately 40 kilograms of freshly harvested banana peduncle samples, which were processed individually to ensure accurate measurement of performance indicators. The study followed standardized testing procedures based on established agricultural machinery evaluation protocols, specifically PAES 235:2008 for multi-crop juice extractors and PAES 229:2005 for fiber decorticators. These standards ensured that the evaluation process was consistent, reliable, and comparable to other agricultural machinery studies.

The performance of the machine was evaluated using multiple response variables, including input capacity, juice extraction rate, juice recovery, extraction losses, extraction efficiency, fuel consumption, fiber extraction rate, fiber recovery, and fiber tensile strength. In addition, the °Brix level of the extracted juice was measured to determine sugar concentration, which is an important factor in ethanol production. The inclusion of both mechanical and biochemical indicators allowed for a comprehensive assessment of the system, covering not only productivity but also product quality. The tensile strength of the extracted fiber was also analyzed using a universal testing machine following ASTM D3822-01 standards to determine whether mechanical processing had adversely affected fiber integrity.

Results of the study showed that both roll rotational speed and crushing load significantly influenced machine performance, although their effects varied depending on the specific output variable. Higher roll rotational speeds generally resulted in increased input capacity, juice extraction rate, and fiber extraction rate due to reduced processing time and faster material throughput within the crushing chamber. Conversely, lower rotational speeds tended to reduce throughput but improved certain recovery metrics due to longer retention time within the extraction system. Crushing load, on the other hand, exhibited a dual effect depending on the response variable. Lower crushing loads facilitated easier passage of peduncle material through the rolls, thereby increasing processing speed, while higher crushing loads enhanced the mechanical compression of the material, resulting in improved juice recovery and extraction efficiency.

The interaction between roll speed and crushing load revealed important trade-offs in system performance. For instance, although higher crushing loads improved juice recovery, they also increased resistance within the system, which reduced processing speed and increased fuel consumption. Similarly, higher rotational speeds improved throughput but contributed to higher extraction losses, as some juice was not fully recovered before the material exited the machine. These findings highlight the complexity of optimizing mechanical systems for biomass processing, where improvements in one performance aspect may negatively affect another. As such, the study emphasized the importance of multi-response optimization techniques to determine the most balanced operating conditions.

Using response surface methodology (RSM), the researchers performed a multi-objective optimization analysis to identify the most desirable combination of operating parameters. The optimization process considered multiple performance goals simultaneously, including maximizing input capacity, juice recovery, extraction efficiency, fiber extraction rate, and fiber recovery, while minimizing extraction losses and fuel consumption. Based on desirability function analysis, the optimal operating condition was identified at approximately 71.378 rpm and 0.8 kN crushing load. At this optimal setting, the machine achieved an input capacity of about 307.24 kg/h, a juice extraction rate of 82.92 kg/h, juice recovery of 26.08%, extraction efficiency of 18.46%, fiber extraction rate of 54.83 kg/h, fiber recovery of 42.29%, and fuel consumption of 0.391 L/h. These results demonstrate that the system can achieve a balanced level of efficiency while maintaining both juice and fiber output quality when operated under optimized conditions.

Further analysis revealed that neither roll rotational speed nor crushing load significantly affected the °Brix content of the extracted juice or the tensile strength of the recovered fiber. This indicates that while the mechanical parameters influence the quantity of juice and fiber produced, they do not significantly alter the chemical composition of the juice or the structural strength of the fiber. This finding is particularly important because it suggests that the quality of the end products remains stable regardless of operational variations, making the system more reliable for industrial applications where consistency is essential.

In terms of fermentation performance, the extracted juice was subjected to ethanol production analysis using Saccharomyces cerevisiae as the fermentation agent. The results showed that the fermented slurry contained approximately 10–11% ethanol by volume across all treatment combinations. Statistical analysis indicated no significant differences in ethanol content among the different mechanical settings, reinforcing the observation that sugar concentration, rather than mechanical extraction conditions, is the primary determinant of ethanol yield. This further supports the conclusion that the machine primarily influences extraction efficiency and throughput rather than biochemical conversion outcomes.

In conclusion, the study demonstrated that the developed banana peduncle juice extractor is an effective machine for converting agricultural waste into valuable products such as bioethanol feedstock and natural fiber. The system successfully addressed the limitations of conventional extraction machines by introducing a design specifically optimized for banana peduncle characteristics. The findings highlight the importance of mechanical optimization in improving productivity while maintaining product quality. Moreover, the study underscores the potential of banana peduncle as a sustainable raw material for bioenergy and fiber-based industries, contributing to waste reduction, renewable energy development, and value-added agricultural processing. The authors further recommended future improvements such as modifying roll geometry, improving transmission efficiency, experimenting with alternative crushing mechanisms, and testing additional banana varieties to enhance the versatility and efficiency of the system.

-A **study by Jiang and Po (2023**) investigated the influence of human resource practices on employee retention in the food manufacturing industry in the Philippines and examined the moderating role of the work environment. The researchers focused on five key human resource practices: training and development, employee participation in decision-making, compensation, job security, and performance appraisal. Grounded in Social Exchange Theory, the study proposed that employees are more likely to remain with an organization when they perceive that the organization provides adequate support, fair treatment, and opportunities for growth. In return, employees reciprocate through commitment, loyalty, and long-term retention.

The study employed a quantitative research design using a cross-sectional survey method. Data were collected from 206 employees working in various food manufacturing companies in the Philippines through both online and offline questionnaires. The researchers used a five-point Likert scale to measure employee perceptions regarding human resource practices, work environment, and employee retention. Statistical analyses, including regression and moderation analysis, were conducted using JAMOVI software to determine the relationships among the variables.

The findings revealed that employee participation in decision-making, compensation, job security, and performance appraisal significantly and positively influenced employee retention. Employees who were given opportunities to participate in organizational decisions tended to develop a stronger sense of belonging and commitment to their companies. Likewise, fair compensation and benefits were found to be major factors that encouraged employees to remain in their organizations. The results further indicated that job security contributed positively to retention because employees preferred stable employment arrangements that reduced uncertainty regarding their future careers. Performance appraisal was also shown to strengthen employee retention by providing feedback, recognition, and opportunities for professional improvement.

Among the identified human resource practices, compensation emerged as the strongest predictor of employee retention. The researchers found that employees were more likely to remain in organizations when they perceived that they were fairly compensated for their work and contributions. This finding emphasizes the importance of providing competitive salaries and benefits in order to attract and retain skilled employees in the food manufacturing sector. The study concluded that compensation plays a critical role in maintaining workforce stability and reducing employee turnover.

In contrast, the study found that training and development did not significantly influence employee retention. This result differed from several previous studies that suggested training opportunities increase employee loyalty and commitment. The researchers proposed that training programs may not automatically lead to retention unless employees perceive them as valuable, relevant, and beneficial to their career advancement. This finding highlights the need for organizations to carefully evaluate the effectiveness of their training and development initiatives and ensure that such programs align with employee needs and organizational objectives.

Another significant contribution of the study was its examination of the work environment as a moderating variable. The findings showed that the work environment strengthened the relationship between employee participation in decision-making, compensation, job security, performance appraisal, and employee retention. Employees working in supportive environments were more likely to respond positively to human resource practices and demonstrate stronger intentions to remain in the organization. The study emphasized that a positive work environment includes both physical and non-physical factors, such as workplace safety, adequate facilities, healthy interpersonal relationships, effective communication, and a supportive organizational culture.

The researchers further explained that improvements in the physical work environment, including proper ventilation, safe production equipment, reduced workplace noise, and clean facilities, can increase employee satisfaction and productivity. At the same time, non-physical aspects such as teamwork, leadership support, organizational structure, and employee relationships contribute significantly to employee motivation and retention. These findings suggest that organizations should not only focus on implementing effective human resource policies but also invest in creating a work environment that promotes employee well-being and engagement.

The study has important implications for food manufacturing companies in the Philippines. It suggests that organizations should prioritize compensation systems, employee involvement, job security measures, and performance evaluation mechanisms to improve retention rates. Furthermore, managers should recognize the importance of maintaining a positive work environment to maximize the effectiveness of these human resource practices. By doing so, organizations can reduce employee turnover, improve workforce stability, and enhance overall organizational performance.

This literature is relevant to the present study because it demonstrates how organizational practices, performance monitoring, and workplace conditions influence efficiency, productivity, and long-term organizational success in the food manufacturing industry. The findings support the development of digital systems that provide real-time monitoring, performance tracking, and data-driven decision-making capabilities. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights the importance of accurate information, continuous monitoring, and effective management practices in improving operational performance and achieving organizational goals.

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### 2.3 Related Literature

[5]A study by **Somsen, Capelle, and Tramper (2004)** introduced Production Yield Analysis (PYA), a systematic method used to evaluate raw material efficiency in food processing. The study developed a "Yield Index" to distinguish between necessary losses and avoidable losses, helping identify inefficiencies in production systems. The authors emphasized that relying solely on historical benchmarks may conceal actual waste, highlighting the importance of model-based analysis. This literature is relevant to the present study as it supports the development of a digital system capable of identifying and analyzing production losses in real time.

[6]A study by **Charbonnier, Béranger, and Cognon (2005)** investigated the application of trend extraction and analysis techniques for complex system monitoring and decision support in industrial and medical environments. The researchers focused on developing an online monitoring methodology capable of analyzing continuously changing data from highly complex

systems such as industrial automation processes, food manufacturing operations, and intensive care monitoring systems. According to the researchers, modern industrial environments generate large volumes of operational data that are difficult for human operators to interpret manually, especially in situations where process conditions rapidly change and immediate decisions are required. The study emphasized that traditional monitoring approaches often rely heavily on manual observation and operator expertise, which may result in delayed responses, reduced efficiency, and increased risk of operational errors during abnormal conditions.

To address these challenges, the researchers proposed a real-time trend extraction methodology designed to analyze univariate time-series data using semi-qualitative representations. The system utilized three primary trend primitives: **Increasing, Decreasing, and Steady**. These primitives were used to represent operational behavior over specific time intervals, allowing users to interpret process conditions more effectively. According to the study, converting numerical signals into symbolic trend representations helps reduce cognitive overload and improves operator understanding of complex operational situations.

The methodology involved several stages. First, continuously generated process data were segmented into linear intervals. These intervals were then classified into temporal patterns such as increasing trends, decreasing trends, stable conditions, transient variations, and step changes. After classification, the detected temporal patterns were transformed into semi-quantitative episodes describing the behavior of operational variables over time. The researchers explained that this approach enabled the monitoring system to identify abnormal process behavior without requiring highly complex mathematical models of the entire industrial system.

The study also explored the application of trend extraction for predictive maintenance and operational supervision in industrial systems. The researchers explained that highly automated production systems often operate under demanding conditions to maximize productivity and profitability, increasing the likelihood of equipment degradation and operational faults. By continuously monitoring operational variables and analyzing abnormal trend patterns, the proposed methodology could assist in identifying equipment deterioration before critical failures occurred. The researchers concluded that predictive monitoring capabilities support preventive maintenance strategies, reduce operational downtime, and improve overall equipment reliability.

[7]A study by **Davies, Fried, and Gather (2003)** entitled “Robust Signal Extraction for On-line Monitoring Data” examined statistical methods for extracting meaningful information from continuously generated time-series data in real-time monitoring environments. The researchers focused on developing and evaluating robust techniques that can effectively separate true underlying signals from noise, outliers, and irregular fluctuations. The study was motivated by the increasing demand for reliable monitoring systems in critical environments such as intensive care units and industrial process control systems, where fast and accurate interpretation of data is essential for decision-making.

The researchers explained that modern monitoring systems generate large volumes of data due to continuous recording of physiological or operational variables. In healthcare settings, for example, patient data such as heart rate, blood pressure, and oxygen levels are recorded at very short intervals, producing complex datasets that must be analyzed immediately. Similarly, industrial systems also generate high-frequency operational data that reflect machine performance, process efficiency, and production status. However, these datasets are often affected by noise, measurement errors, and external disturbances, which make it difficult to identify meaningful patterns using traditional analysis methods.

According to the study, one of the major challenges in online monitoring is distinguishing clinically or operationally relevant changes from irrelevant noise and outliers. The researchers emphasized that false alarms in monitoring systems are a significant problem, particularly in intensive care environments where excessive alarms may lead to alarm fatigue among medical staff. This reduces trust in automated systems and may cause important warnings to be ignored. In industrial environments, inaccurate signal interpretation may also lead to unnecessary maintenance actions or failure to detect real system faults. Because of these issues, there is a strong need for robust statistical methods that can reliably extract true signals from noisy data in real time.

Another important contribution of the study is its discussion on the trade-off between robustness and computational efficiency. The researchers emphasized that while highly robust methods provide better resistance to data contamination, they may not always be suitable for real-time applications due to processing delays. Therefore, selecting an appropriate signal extraction method requires balancing accuracy, stability, and computational cost depending on the system requirements.

In conclusion, the study by Davies, Fried, and Gather (2003) provides valuable insights into the importance of robust statistical methods in real-time monitoring systems. Its emphasis on noise reduction, trend extraction, and reliable signal interpretation makes it highly applicable to modern industrial systems, particularly in the context of automated food production and digital manufacturing technologies.

[8]A study conducted by **Charmine Sheena R. Saflor and Marvin I. Noroña (2021)** explored the implementation of Total Productive Maintenance (TPM) within the Philippine rice production sector. The research focused on addressing the increasing operational and maintenance challenges experienced by farmers and agricultural cooperatives due to the growing use of mechanized farming equipment. The authors emphasized that many rice production operators in the Philippines still rely on reactive maintenance practices, where machines are repaired only after breakdowns occur. This approach often results in frequent equipment failures, increased maintenance expenses, production delays, and reduced overall productivity. The study highlighted that inadequate maintenance management can shorten the lifespan of agricultural machinery and negatively affect the efficiency of rice production operations.

The study is relevant to the present research because it demonstrates the importance of monitoring equipment performance and maintenance activities to improve operational efficiency and reduce production losses. It supports the idea that implementing a digital and data-driven monitoring system can help organizations identify inefficiencies, manage maintenance schedules, and enhance productivity in real time. Similar to the proposed study on real-time yield analytics and extraction efficiency monitoring, the research emphasizes the value of systematic analysis and continuous monitoring in optimizing production processes, minimizing waste, and improving decision-making within industrial and agricultural operations.

[9]A study conducted by **Desai Shivani Sarjerao (2020),** Design, Development and Performance Evaluation of Juice Extractor, presented the development of a hydraulically operated juice extraction machine intended for processing pomegranate arils and bottle gourd.

The experimental evaluation focused on three main independent variables: size reduction (shred size), applied hydraulic pressure, and filter bag pore size. These were tested in different combinations to determine their effects on several dependent performance indicators, namely juice yield, juice extraction efficiency, extraction losses, machine capacity, and sedimentation percentage. A series of experimental trials were conducted for both pomegranate arils and bottle gourd, with repeated measurements used to ensure reliability of results. Statistical analysis, including ANOVA, was applied to determine the significance of main effects and interaction effects among the variables.

In terms of juice extraction efficiency, the study reported maximum values of 94.30% for pomegranate and 80.27% for bottle gourd under optimal operating conditions. The results also showed a consistent decrease in extraction losses as pressure increased and shred size decreased, demonstrating improved material utilization and reduced wastage. Although most interaction effects between variables were statistically non-significant, individual factors such as pressure and filter bag pore size were found to have a significant impact on extraction performance.

Overall, the study emphasizes that mechanical design parameters and operating conditions play a critical role in determining the efficiency of juice extraction systems. It further suggests that optimizing these variables can significantly enhance juice recovery, reduce processing losses, and improve machine productivity. This research is relevant to the present study as it provides an engineering-based framework for evaluating extraction systems and supports the development of optimized, efficiency-driven food processing technologies in agricultural production systems.

## 2.4 SYNTHESIS OF THE REVIEWED LITERATURE

**-Summary of Findings**

The collection of foreign and local studies reviewed in this chapter consistently highlights the increasing importance of digital transformation in modern production and manufacturing environments. Across various industries, researchers have emphasized that organizations are moving away from traditional paper-based methods and adopting digital technologies to improve operational efficiency, data accuracy, resource utilization, and decision-making processes. The reviewed literature demonstrates that digitalization has become a critical strategy for organizations seeking to remain competitive in an increasingly technology-driven environment.

One of the major themes identified from the reviewed studies is the significance of real-time data monitoring and management. Verhoef et al. (2021) explained that conventional manual recording systems often create information gaps that prevent managers from detecting production issues immediately. Because production data are recorded and reviewed only after operations have been completed, organizations frequently experience delays in identifying inefficiencies, wastage, and equipment-related problems. The researchers found that centralized digital systems provide immediate access to operational information, allowing managers to respond more effectively to changing production conditions. This finding reinforces the idea that timely access to accurate information is essential for maintaining production efficiency and reducing operational losses.

Similarly, Makhmudah et al. (2025) demonstrated that implementing digital production reporting systems can significantly improve data accuracy and operational efficiency. Their findings revealed that replacing manual reporting methods with automated and centralized systems reduces the likelihood of human error while improving the speed of information processing. The study further showed that digital dashboards and automated reporting tools provide management with better visibility into operational performance, enabling more informed and timely decision-making. These findings suggest that digital monitoring systems play an important role in enhancing organizational productivity and supporting effective production management.

The reviewed local studies also support the growing need for digital transformation within organizations. Bagaforo and Albason (2024) examined the adoption of digital technologies among Micro, Small, and Medium Enterprises (MSMEs) in the Philippines and found that digitalization contributes significantly to business competitiveness, operational effectiveness, and resource optimization. Their study revealed that organizations that successfully integrate digital technologies into their operations gain advantages in terms of efficiency, responsiveness, and overall performance. Although challenges related to technological readiness and infrastructure remain, the researchers emphasized that digital transformation offers substantial benefits for organizations seeking to improve operational processes.

Likewise, the study conducted by De Leon et al. (2017) demonstrated the value of centralized information systems in improving organizational operations. By implementing a database-driven management system, the researchers were able to improve data accessibility, security, and reliability while reducing dependence on manual record-keeping practices. Their findings support the argument that centralized digital platforms can effectively address common issues associated with paper-based systems, including data loss, inconsistency, and inefficiency.

Another significant theme that emerged from the reviewed literature is the importance of production yield analysis and resource utilization monitoring. Somsen, Capelle, and Tramper (2004) introduced the concept of Production Yield Analysis as a systematic approach to evaluating production efficiency and identifying avoidable losses within food processing operations. Their study emphasized that organizations often underestimate production losses when relying solely on traditional performance indicators or historical benchmarks. Through detailed yield analysis, organizations can identify specific areas where resources are being wasted and implement corrective measures to improve efficiency. This finding highlights the importance of continuously monitoring production performance to maximize resource utilization and reduce unnecessary losses.

The literature also demonstrates the growing relevance of data analytics and trend analysis in modern production environments. Charbonnier, Béranger, and Cognon (2005) emphasized that industrial systems generate large volumes of operational data that are often difficult to interpret using conventional monitoring methods. Their research showed that trend extraction techniques can simplify complex datasets and help operators identify abnormal process behavior in real time. By transforming raw operational data into meaningful patterns, organizations can improve process visibility, enhance operational control, and respond more effectively to emerging production issues.

Supporting this perspective, Davies, Fried, and Gather (2003) highlighted the importance of extracting meaningful signals from continuously generated monitoring data. The researchers found that monitoring systems frequently encounter noise, irregularities, and data fluctuations that may obscure critical operational information. Their study demonstrated that robust signal extraction methods improve the reliability of monitoring systems by distinguishing actual process changes from irrelevant variations. The findings suggest that effective monitoring systems should not only collect data but also provide mechanisms for filtering, analyzing, and interpreting information accurately.

The reviewed studies further emphasize the importance of monitoring equipment performance and maintenance activities. Saflor and Noroña (2021) investigated the implementation of Total Productive Maintenance in the Philippine rice production sector and found that systematic monitoring and preventive maintenance practices significantly improve operational efficiency and equipment reliability. Their findings revealed that organizations that actively monitor machine performance experience fewer equipment failures, reduced downtime, and lower maintenance costs. This demonstrates that continuous monitoring is essential not only for production output but also for maintaining the overall effectiveness of production equipment.

In addition, the study conducted by Desai (2020) provided valuable insights into factors affecting juice extraction performance. The researcher demonstrated that operational variables such as hydraulic pressure, shred size, and filter characteristics significantly influence extraction efficiency, juice yield, and production losses. The study found that optimizing these variables can substantially improve resource utilization and production outcomes. These findings are particularly relevant to the present study because they highlight the importance of continuously monitoring extraction performance to maximize yield and minimize losses during food processing operations.

Collectively, the reviewed literature and studies establish a strong theoretical and practical foundation for the development of digital production monitoring systems. They consistently demonstrate that organizations benefit from real-time monitoring, centralized data management, automated analytics, and continuous performance evaluation. Furthermore, the findings indicate that monitoring production yield and extraction efficiency can significantly contribute to operational improvement, waste reduction, and better decision-making. These concepts directly support the objectives of the present study, which seeks to develop a digital production monitoring system that provides real-time yield analytics and extraction efficiency monitoring for AIDi Foods Inc.-

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**Research Gaps Identified**

Despite the substantial body of literature related to digital transformation, production monitoring, data analytics, and operational efficiency, several gaps remain evident within existing research. These gaps provide justification for the development of the proposed study and highlight areas where further investigation is necessary.

One major gap identified from the reviewed studies is the limited focus on food manufacturing operations, particularly those involving juice extraction processes. While many researchers have explored digital transformation and real-time monitoring in general manufacturing environments, relatively few studies have specifically examined how digital technologies can be applied to monitor extraction efficiency and yield performance within food processing facilities. Most existing studies focus on broader production management systems rather than specialized monitoring of extraction-related performance indicators.

Another significant gap involves the integration of yield analytics and extraction efficiency monitoring within a single digital platform. The reviewed studies often address production monitoring, maintenance management, data reporting, or process optimization independently. However, there is limited research that combines these elements into a unified system capable of simultaneously tracking raw material inputs, production outputs, yield percentages, extraction losses, and operational efficiency in real time. As a result, organizations may still rely on multiple disconnected systems or manual procedures to evaluate production performance.

A further gap relates to the technological requirements of many existing monitoring solutions. Several studies examined systems that rely heavily on sophisticated industrial automation technologies, sensor networks, Industrial Internet of Things (IIoT) infrastructures, and advanced machine learning algorithms. While these technologies offer substantial benefits, they may not be practical or financially feasible for many small and medium-sized enterprises. Consequently, there is a need for more accessible and cost-effective digital monitoring solutions that can deliver meaningful operational insights without requiring extensive automation infrastructure.

The reviewed literature also revealed a shortage of localized studies focusing specifically on production monitoring within Philippine food manufacturing organizations. Although local studies have explored digital transformation, business information systems, and organizational technology adoption, very few have investigated the implementation of real-time production analytics systems in food processing environments. This creates a research gap concerning the applicability of digital production monitoring technologies within local manufacturing settings and highlights the need for studies that address the operational realities of Philippine-based enterprises.

Furthermore, many of the reviewed studies concentrated primarily on improving data collection and reporting processes but provided limited discussion regarding how collected data can be transformed into actionable insights for decision-making. While data acquisition remains important, organizations increasingly require systems capable of converting raw operational information into meaningful analytics that support productivity improvement, waste reduction, and strategic planning. This indicates a need for monitoring systems that not only gather information but also provide analytical capabilities that facilitate informed managerial decisions.

Finally, existing studies rarely focus on the specific challenges faced by organizations that continue to rely on manual production recording systems. Many researchers assume the existence of some level of digital infrastructure, leaving a gap in understanding how organizations can effectively transition from paper-based operations toward digital production management. This limitation is particularly relevant for organizations such as AlDi Foods Inc., where manual data recording remains a significant operational challenge.

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**How the Current Project Addresses the Gaps**

The proposed study directly addresses the identified research gaps through the design and development of a Digital Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System specifically intended for AlDi Foods Inc. Unlike many existing studies that focus on general manufacturing environments, the proposed system is tailored to the operational requirements of food production processes, particularly juice extraction operations where yield and extraction efficiency are critical performance indicators.

The system addresses the lack of specialized monitoring solutions by integrating production monitoring, yield analytics, and extraction efficiency evaluation within a single centralized platform. Through automated computation and real-time data processing, the system enables users to monitor raw material consumption, production outputs, yield percentages, and extraction performance simultaneously. This integrated approach eliminates the need for fragmented monitoring processes and provides a more comprehensive view of operational performance.

The proposed system also addresses the challenge of limited real-time visibility by replacing manual production logs with a centralized digital database and dashboard. Through real-time monitoring and visualization features, production managers can immediately identify production inefficiencies, abnormal performance trends, and potential extraction losses as they occur. This capability supports proactive decision-making and allows corrective actions to be implemented before significant losses accumulate.

In addition, the study offers a practical and cost-effective alternative to highly sophisticated industrial automation systems. Unlike solutions that require extensive IIoT infrastructures or advanced machine-learning technologies, the proposed system focuses on implementing accessible digital tools that can be realistically adopted by small and medium-sized enterprises. This makes the solution more feasible for organizations seeking to begin their digital transformation journey without incurring substantial technological investment costs.

The proposed research also contributes to the limited body of local knowledge regarding digital production monitoring within the Philippine food manufacturing sector. By focusing on the operational environment of AlDi Foods Inc., the study provides practical insights into how local manufacturing organizations can utilize digital technologies to improve production monitoring, resource utilization, and operational efficiency.

Ultimately, the proposed system seeks to bridge the gap between traditional manual production management practices and modern data-driven manufacturing approaches. By combining centralized data management, real-time monitoring, automated analytics, yield evaluation, and extraction efficiency assessment, the study provides a comprehensive solution that supports productivity improvement, waste reduction, and evidence-based decision-making within food production operations. This contribution aligns with the objectives of Industry 4.0 and supports the continued digital transformation of manufacturing organizations.

# CHAPTER 3

# METHODOLOGY

## -3.1 Research Design

This study will employ a **Developmental Research Design** to guide the systematic analysis, design, development, implementation, and evaluation of the proposed system entitled Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc. Developmental research is an appropriate methodology for information technology projects because it focuses on creating practical solutions to existing organizational problems through the development of software systems and technological innovations.

The primary objective of this research is to develop a digital production monitoring platform capable of replacing the existing manual and paper-based recording processes currently utilized by AlDi Foods Inc. The proposed system aims to provide real-time monitoring of production activities, automated yield calculations, extraction efficiency analysis, production stability interpretation, and periodic raw material usage reporting. Through the implementation of these features, the organization will be able to monitor production performance more effectively and make data-driven decisions based on accurate and timely information.

The developmental research design follows a systematic process that begins with identifying existing operational challenges through data gathering activities such as interviews, observations, and document reviews. Information gathered from these activities will serve as the basis for determining system requirements and identifying opportunities for process improvement. The findings will then be translated into system specifications that address the company's production monitoring needs.

Furthermore, the study emphasizes the iterative process of software development wherein continuous testing, validation, and refinement are conducted throughout the project lifecycle. Each component of the proposed system will undergo evaluation to ensure that it meets functional requirements and aligns with the operational objectives of AlDi Foods Inc. The iterative nature of developmental research allows the researchers to continuously improve the system based on feedback obtained during development and testing phases.

The proposed system will specifically focus on monitoring raw material inputs, extraction outputs, packed yield records, yield percentages, extraction efficiency indicators, and production stability status. Additionally, the system will generate analytical reports that summarize raw material utilization on a weekly, monthly, quarterly, and yearly basis. These functionalities are intended to provide management with valuable insights into production performance and resource utilization trends.

Through this research design, the study seeks to produce a functional and reliable digital monitoring platform that contributes to improved operational efficiency, reduced production losses, enhanced reporting accuracy, and better managerial decision-making within the organization.

### 3.2 System Development Methodology

The study will utilize the **Agile Development Model** as the primary software development methodology. Agile is an iterative and flexible approach to software development that emphasizes continuous improvement, stakeholder collaboration, rapid delivery of functional components, and adaptability to changing requirements. This methodology is particularly suitable for the proposed system because it allows developers to continuously refine features based on user feedback and evolving organizational needs.

Unlike traditional software development approaches that follow a rigid sequential structure, Agile divides the project into smaller and manageable development cycles known as sprints. Each sprint focuses on a specific set of functionalities that are designed, developed, tested, and evaluated before proceeding to the next phase. This iterative approach enables the researchers to identify and resolve issues early in the development process while ensuring that each system component meets the required quality standards.-

The Agile Development Model for this study consists of the following phases:

#### Requirements Analysis

The first phase involves gathering and analyzing information regarding the existing production monitoring process of AlDi Foods Inc. Data collection activities will include interviews with production personnel, direct observation of operational workflows, and examination of existing production records and reports.

The primary objective of this phase is to identify operational challenges associated with manual production recording, determine user requirements, and establish the functional and non-functional requirements of the proposed system. Particular attention will be given to understanding how production data is currently collected, monitored, and reported, as well as identifying opportunities for automation and process improvement.

**System Design**

Following the completion of the requirements analysis phase, the researchers will create the overall design of the proposed system. This includes designing the system architecture, database structure, user interface layouts, process flow diagrams, and system navigation framework.

The database design will focus on storing and managing production-related information such as user accounts, production batches, raw material inputs, extraction outputs, packed yields, yield percentages, extraction efficiency records, and historical analytics data. User interface designs will prioritize simplicity, accessibility, and ease of use to ensure efficient interaction between users and the system.

Additionally, system workflows will be designed to support real-time monitoring, automated calculations, report generation, and dashboard visualization functionalities.

**Development**

During the development phase, the proposed system will be built using the selected technologies and programming tools. The frontend interface will be developed using HTML and CSS, while PHP will be utilized for backend processing and business logic implementation. MySQL will serve as the centralized database management system, -and Chart.js will be used to generate graphical analytics and dashboard visualizations.

Development activities will focus on implementing the major modules of the system, including:

- User Authentication and Access Management
- Production Management Module
- Raw Material Recording Module
- Extraction Monitoring Module
- Packed Yield Monitoring Module
- Real-Time Yield Analytics Module
- Extraction Efficiency Monitoring Module
- Production Stability Interpretation Module
- Historical Data Management Module
- Dashboard and Visualization Module

- Report Generation Module

The production stability interpretation module will automatically evaluate current production performance and classify operational status as either **Stable** or **Not Stable** based on predefined yield efficiency thresholds established by the organization.

-Furthermore, the reporting module will generate periodic raw material usage reports categorized into weekly, monthly, quarterly, and yearly summaries. These reports are intended to support production planning, resource management, and long-term operational analysis.-

**Testing**

After development, the system will undergo comprehensive testing procedures to ensure that all functionalities operate as intended. Testing activities will be conducted throughout the development process to identify and correct errors before deployment.

-The testing process will include:

**Unit Testing** – Individual system components and modules will be tested independently to verify their functionality and accuracy.

**System Testing –** The entire system will be tested as an integrated application to ensure proper interaction among all modules.

**User Acceptance Testing (UAT)** – Selected users from AlDi Foods Inc. will evaluate the system to determine whether it satisfies operational requirements and user expectations.

Special attention will be given to validating the accuracy of yield calculations, extraction efficiency computations, production stability interpretations, and periodic report generation functions.-

**Deployment (Prototype)**

Upon successful completion of testing, the proposed system will be deployed as a prototype within a controlled environment for demonstration and evaluation purposes. The deployment phase allows stakeholders to experience the actual operation of the system and assess its effectiveness in addressing existing production monitoring challenges.

The prototype implementation will demonstrate the system's capability to perform real-time production monitoring, generate automated analytics, display production dashboards, classify operational stability, and produce periodic raw material utilization reports.

Feedback obtained during this phase will be utilized to identify additional improvements and enhancements prior to future implementation or expansion of the system.

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**3.3 System Architecture**

The proposed system will utilize a classic three-tier architecture, which systematically divides the platform's core infrastructure into a Client Layer, an Application Layer, and a Database Layer. This architectural design promotes highly efficient communication between different system components while keeping their core functions separate. By isolating user interaction, business logic, and data storage into distinct levels, the system inherently ensures long-term scalability, easier code maintainability, and robust data integrity during high-volume operations.

This structured three-tier layout is specifically engineered to handle the demanding, real-time monitoring and complex analytical requirements of AlDi Foods Inc.'s production operations. Because the Application Layer sits independently between the user interface and the storage engine, it can process heavy data streams from the factory floor—such as live yield calculations and extraction metrics—without slowing down the user experience. This ensures that management receives lag-free, uninterrupted operational insights while maintaining a highly stable and secure digital environment.

**Client Layer**

The Client Layer serves as the primary interface between users and the system. This layer allows production personnel, supervisors, and administrators to interact with the application through a web-based platform. Users can input production-related information such as raw material quantities, extraction output measurements, packed yield records, and other operational data necessary for monitoring production performance.

In addition to data entry, the Client Layer provides access to the system's centralized dashboard, where users can view real-time production statistics and analytical reports. Through graphical charts, tables, and visual indicators, users can monitor current yield percentages, extraction efficiency levels, production trends, and raw material utilization. The dashboard also displays automated production status interpretations, allowing users to immediately determine whether a production batch is classified as Stable or Not Stable.

The user interface is designed to be intuitive, responsive, and easy to navigate, ensuring that production personnel can efficiently perform their tasks without requiring extensive technical knowledge. By providing real-time access to production information, the Client Layer enhances operational visibility and supports faster decision-making.

**Application Layer**

The Application Layer functions as the core processing component of the system. It is responsible for executing all business logic, system rules, analytical computations, and data processing activities. This layer receives information submitted by users through the Client Layer and transforms it into meaningful operational insights.

One of the primary responsibilities of the Application Layer is performing automated calculations related to production monitoring. These calculations include yield percentage computation, extraction efficiency analysis, and raw material utilization tracking. The system continuously processes incoming production data to generate updated performance metrics that can be viewed in real time through the dashboard.

Furthermore, the Application Layer contains the Production Stability Interpretation Module. This module evaluates production performance based on predefined operational thresholds established by the organization. Using the calculated yield and extraction efficiency values, the system automatically classifies production performance as either Stable or Not Stable. This automated interpretation helps management identify operational issues early and supports immediate corrective action when necessary.

The Application Layer is also responsible for generating periodic analytical reports. Using stored production records, the system automatically prepares weekly, monthly, quarterly, and yearly raw material usage reports. These reports provide management with a comprehensive overview of resource consumption patterns, production efficiency trends, and operational performance over time.

In addition, this layer manages user authentication, access control, report generation, historical record retrieval, and dashboard visualization processes. By centralizing all processing activities within the Application Layer, the system ensures consistency, accuracy, and reliability across all operational functions.

**Database Layer**

The Database Layer serves as the centralized repository for all system information. It is responsible for storing, organizing, retrieving, and securing production data generated throughout the manufacturing process. The database ensures that all records remain accessible, accurate, and protected from unauthorized modification.

The database stores various categories of information, including user accounts, production batch records, raw material inputs, extraction outputs, packed yield records, yield percentages, extraction efficiency values, stability classifications, and generated analytical reports. Historical production records are also maintained to support long-term trend analysis and performance evaluation.

By maintaining a centralized database, the system eliminates the risks associated with paper-based record keeping, such as data loss, duplication, inconsistency, and delayed retrieval. Production information can be accessed instantly whenever needed, allowing management to generate reports, review historical performance, and monitor operational efficiency more effectively.

Furthermore, the database supports the system's reporting and analytics functions by providing structured and organized data for processing. Information stored within the database serves as the foundation for generating real-time dashboards, production summaries, raw material utilization reports, and extraction efficiency analyses.

**System Data Flow**

The operation of the system begins when authorized users enter production-related information through the Client Layer. The submitted data is transmitted to the Application Layer, where validation, processing, and analytical computations are performed. The processed information is then stored within the Database Layer for future reference and analysis.

After processing, the Application Layer retrieves relevant information from the database and presents the results back to users through the dashboard interface. Users can then view real-time yield percentages, extraction efficiency measurements, production stability status, historical records, and periodic raw material usage reports. This continuous flow of information enables proactive monitoring, supports data-driven decision-making, and contributes to improved production efficiency within AlDi Foods Inc.

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### 3.4 Tools and Technologies Used

The development of the proposed system requires the strategic integration of various software technologies and specialized development tools. This carefully selected tech stack is essential to ensure high-efficient system performance, highly accurate data processing, and robust real-time monitoring capabilities on the production floor. Additionally, these tools will provide reliable database management, ensuring that all critical manufacturing data is securely stored, organized, and retrieved without lag.

Every selected technology was meticulously chosen based on its compatibility, accessibility, ease of development, and overall suitability for web-based information systems. By prioritizing tools that seamlessly connect with one another, the development team can minimize integration roadblocks and accelerate the deployment process. Ultimately, this tailored combination of software ensures that the final platform is both highly stable for developers to maintain and easily accessible for users across the organization.

**Frontend Technologies**

**HTML (HyperText Markup Language)**

HTML will serve as the primary markup language used in developing the user interface of the proposed system. It provides the structural foundation of the web application by organizing content, forms, tables, navigation menus, dashboards, and system pages.

Through HTML, users will be able to interact with various system functionalities such as entering raw material data, recording extraction outputs, monitoring production batches, viewing yield analytics, and generating production reports. The use of HTML ensures compatibility across different web browsers and devices, allowing users to access the system through a standard web interface.

**CSS (Cascading Style Sheets)**

CSS will be utilized to enhance the visual appearance and usability of the system. It controls the layout, formatting, colors, typography, and responsiveness of the user interface.

The implementation of CSS improves user experience by creating a clean and organized dashboard environment where production managers and operators can easily access production information. Responsive design techniques will also be incorporated to ensure that the system remains functional and visually consistent across various screen sizes and devices.

**Backend Technology**

**PHP (Hypertext Preprocessor)**

PHP will be used as the primary server-side programming language responsible for handling business logic, data processing, and communication between the user interface and the database.

The proposed system will utilize PHP to perform several critical functions, including:

- User authentication and access control
- Production data processing
- Yield percentage calculation
- Extraction efficiency computation
- Production stability interpretation
- Report generation
- Historical data retrieval
- Dashboard data processing

PHP enables the system to process production records dynamically and generate real-time analytical outputs based on user inputs. The language is particularly suitable for database-driven web applications due to its strong integration capabilities with MySQL.

**-Analytics Processing Technology**

**<u>Chart.js</u>**

Chart.js will be utilized as the primary data visualization library for displaying production analytics and operational reports. The library provides interactive graphical representations of production data, enabling users to understand performance trends quickly and effectively.

The system will use Chart.js to visualize:

- Current Yield Percentage
- Extraction Efficiency Performance
- Stable and Not Stable Production Trends
- Raw Material Consumption Analysis
- Production Output Trends
- Weekly Usage Reports
- Monthly Usage Reports
- Quarterly Usage Reports
- Yearly Usage Reports

By converting raw numerical information into visual charts and graphs, Chart.js improves data interpretation and supports data-driven decision-making within AlDi Foods Inc.-

**Database Technology**

**MySQL**

MySQL will serve as the centralized database management system of the proposed application. It is responsible for storing, organizing, retrieving, and managing all production-related information generated by the system.

The database will contain records such as:

- User Accounts
- Production Batches
- Raw Material Inputs
- Extraction Outputs
- Packed Yield Records
- Yield Percentage Results
- Extraction Efficiency Results
- Production Stability Status
- Historical Production Records
- Generated Reports

MySQL provides efficient data storage and retrieval capabilities, ensuring that production information remains accurate, secure, and accessible whenever required. The centralized database also supports historical analysis by preserving production records over extended periods.

### Development Environment

### Visual Studio Code

Visual Studio Code (VS Code) will be utilized as the primary Integrated Development Environment (IDE) for developing the proposed system. It provides a comprehensive coding environment that supports frontend and backend development activities.

Visual Studio Code offers several features that improve development efficiency, including:

- Syntax highlighting
- Code completion
- Debugging tools
- Extension support
- Integrated terminal
- Version control integration

These features enable developers to write, test, and maintain system code efficiently throughout the development lifecycle.

### Local Development Server

### XAMPP

XAMPP will be used as the local development server environment during system development and testing. It provides the necessary components required for executing PHP applications and managing MySQL databases within a local machine.

XAMPP includes:

- Apache Web Server
- MySQL Database Server
- PHP Interpreter
- phpMyAdmin Database Management Tool

The use of XAMPP allows the researchers to develop, test, and validate system functionalities before deployment.

**-Technology Integration**

The proposed system operates through the integration of these technologies. HTML and CSS provide the user interface through which production personnel interact with the system. PHP processes incoming production data and performs calculations related to yield analytics and extraction efficiency monitoring. MySQL stores production records and historical information, while Chart.js converts processed data into graphical dashboards and analytical reports. Visual Studio Code and XAMPP serve as the development and testing environment that supports the creation and validation of the entire application.

Together, these technologies provide the technical foundation necessary to implement a digital production monitoring platform capable of supporting real-time yield analytics, extraction efficiency monitoring, production stability interpretation, and periodic raw material usage reporting for AIDi Foods Inc.

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### 3.5 Data Gathering Techniques

To obtain the necessary information for the design and development of the proposed system, the researchers will deploy a strategic mix of data gathering techniques. Rather than relying on a single source of information, this multi-layered approach ensures that data is collected from various operational perspectives within the facility. This comprehensive strategy forms the vital first step of the entire research and development lifecycle.

These chosen data gathering methods will provide the team with a deep, comprehensive understanding of the existing production monitoring process at AIDi Foods Inc. By mapping out how day-to-day logs are currently handled, the researchers can trace the movement of information from the initial processing stages to final packaging. Gaining this clear overview of current floor habits ensures the technical team builds on a realistic foundation.

Furthermore, this data gathering process is specifically tailored to identify deep-seated operational challenges and bottlenecks that plague the current workflow. Whether the issues stem from delayed paper reports, data entry typos, or siloed spreadsheets, capturing these pain points allows the researchers to determine the precise functional requirements of the proposed digital system. In short, it highlights exactly what problems the new software needs to solve.

Ultimately, this thorough data gathering phase is essential to ensure that the final developed system accurately and directly addresses the unique needs of the organization. By aligning the software design with real-world factory requirements, the platform will be fully equipped to support critical features like real-time yield analytics, extraction efficiency monitoring, and automated production reporting. This guarantees that the final digital solution delivers immediate, practical value to management and floor staff alike.

**Interviews**

The researchers will conduct interviews with selected personnel of AlDi Foods Inc., including production managers, supervisors, operators, and other individuals directly involved in the production process. The interviews aim to gather firsthand information regarding the current methods of recording, monitoring, and reporting production data.

Through these interviews, the researchers will identify the strengths and limitations of the existing manual recording system, understand how production information is currently processed, and determine the challenges encountered during daily operations. Particular attention will be given to issues related to data accuracy, delayed reporting, difficulty in monitoring yield performance, extraction losses, and decision-making processes.

The interview process will also allow stakeholders to express their expectations and recommendations regarding the proposed system. Information gathered from the interviews will serve as a primary source for identifying system requirements and determining the features necessary to improve production monitoring and operational efficiency.

### Observation

Direct observation will be conducted within the production environment to examine the actual workflow and operational procedures of AIDi Foods Inc. This technique allows the researchers to gain a better understanding of how production activities are performed and how data is collected during manufacturing operations.

During the observation process, the researchers will examine the procedures involved in recording raw material inputs, monitoring extraction outputs, documenting packed yields, and generating production reports. The researchers will also observe how production personnel manage data recording tasks and identify areas where inefficiencies, delays, or errors commonly occur.

Observation provides valuable information that may not be fully captured through interviews alone. By witnessing the production process firsthand, the researchers can verify existing procedures, identify workflow bottlenecks, and determine opportunities for process automation and digitalization.

Furthermore, observation will help the researchers understand how real-time monitoring capabilities can be integrated into existing operations without significantly disrupting established production workflows.

**Document Review**

The researchers will conduct a thorough review of existing documents, records, and production reports utilized by AlDi Foods Inc. Document review serves as an important source of factual and historical information regarding production activities and reporting practices.

The documents to be reviewed may include:

- Production Logs
- Daily Production Reports
- Raw Material Usage Records
- Extraction Output Records
- Packed Yield Reports
- Inventory Records
- Historical Production Data
- Existing Monitoring Forms

The primary purpose of reviewing the company's existing monitoring forms is to understand exactly how production information is currently organized, recorded, and analyzed within the organization. By conducting a thorough examination of these physical or digital documents, the researchers can map out the current state of data management. This process allows the team to see how information is categorized on a day-to-day basis and how frontline workers interact with logging tools.

During this review, the researchers will closely examine the structure of existing records to identify the specific data elements required for monitoring production performance. This involves tracking how information flows throughout the production process, from the initial raw material staging to the final output. Pinpointing these critical data touchpoints ensures that no vital variables are missed when transitioning from the old system to the new digital platform.

Additionally, analyzing historical production records will provide valuable insights into long-term production trends, material utilization patterns, yield performance, and extraction efficiency measurements. Looking at past data helps the team understand normal operational baselines and recurring bottlenecks. Ultimately, these findings will establish the precise data requirements for the proposed system, directly supporting the development of automated reporting and advanced analytical functionalities.

**Summary of Data Gathering Process**

To thoroughly understand the operational landscape, researchers will utilize a strategic combination of interviews, direct observation, and document reviews. This multi-faceted approach allows the team to gather detailed, firsthand insights from different layers of the company. By engaging directly with staff and analyzing physical workflows, the researchers can build a complete and realistic picture of how data currently moves through the facility.

The primary goal of gathering this comprehensive information is to deeply analyze the current production monitoring practices of AlDi Foods Inc. By examining existing habits, legacy spreadsheets, and floor behaviors, the team can accurately identify underlying operational challenges and bottlenecks. Uncovering these pain points is a critical step, as it highlights exactly where the current manual methods are falling short or causing delays.

Once these challenges are clearly understood, the gathered data will be used to define precise system requirements and design robust database structures. The researchers will translate real-world floor needs into technical blueprints, ensuring the underlying architecture can handle the company's data volume safely. This foundational design work ensures that the software is tailor-made to fit the unique operational realities of the plant.

Furthermore, these insights will directly guide the development of specific monitoring functionalities and advanced analytical tools. The software will be engineered to support critical factory needs, such as real-time yield analytics and extraction efficiency monitoring. By building tools based on actual factory metrics, the resulting platform will be fully equipped to track raw material transformations precisely as they happen.

Ultimately, the collected data will serve as the indispensable foundation for the development of the entire digital production monitoring system. The final platform is engineered to transform company operations by drastically improving data accuracy, enhancing live operational visibility, and reducing tedious manual recording efforts. By replacing paper logs with automated tracking, the system will successfully support data-driven decision-making across the entire organization.

### 3.6 Testing Procedures

The proposed system will undergo a series of structured testing procedures before its official deployment at AlDi Foods Inc. This rigorous evaluation phase is designed to ensure that all developed functionalities operate correctly, efficiently, and reliably under real-world conditions. By systematically walking through each feature in a controlled environment, the project team can guarantee that the platform is robust, stable, and completely safe to integrate into daily operations.

These thorough testing procedures are essential to validate the system's capability to handle complex, data-driven manufacturing workflows. Specifically, the technical team will test the platform's ability to support real-time yield analytics and extraction efficiency monitoring. Ensuring these features work flawlessly means that the data flowing from the production floor is captured accurately and processed without delay.

Furthermore, the testing process will focus heavily on the system's advanced analytical capabilities, such as automated production stability interpretation and periodic raw material usage reporting. The software must be able to evaluate production trends automatically and generate precise, dependable summaries of resource consumption. Testing these components ensures that management receives highly accurate insights to help optimize resources and eliminate waste.

Ultimately, the primary aim of this comprehensive testing process is to identify hidden software errors, eliminate system bugs, and verify overall performance. By matching the system's actual behavior against its original design blueprints, the team can confidently ensure that the developed platform meets all functional and non-functional requirements of the study. This final validation guarantees a high-quality digital solution that aligns perfectly with the company's operational goals.

**-Unit Testing**

Unit testing will be conducted to evaluate individual components or modules of the system independently. Each function will be tested to ensure that it performs its intended task accurately before being integrated into the full system.

The following modules will be subjected to unit testing:

- User authentication and login module
- Raw material input module
- Extraction output and packed yield input module
- Yield percentage computation module
- Extraction efficiency calculation module
- Production stability interpretation module (Stable / Not Stable classification)
- Dashboard data retrieval module
- Report generation module (weekly, monthly, quarterly, yearly)

During unit testing, the researchers will verify the correctness of calculations, particularly the accuracy of yield percentage computation and extraction efficiency results. The stability classification logic will also be tested to ensure that production status is correctly determined based on predefined thresholds.

### System Testing

System testing will be performed to evaluate the complete and integrated system as a whole. This stage ensures that all modules work together properly and that data flows correctly across the Client, Application, and Database layers.

The system will be tested under simulated production scenarios that reflect the actual workflow of AlDi Foods Inc., including:

- Input of raw material quantities during production
- Recording of extraction and packed yield outputs
- Real-time computation of yield percentage
- Continuous monitoring of extraction efficiency
- Automatic classification of production stability
- Generation of dashboard analytics and visual reports
- Creation of periodic raw material usage reports

System testing will ensure that all components interact seamlessly and that real-time updates are reflected accurately on the dashboard without data loss, delay, or inconsistency.

### User Acceptance Testing (UAT)

User Acceptance Testing will be conducted with selected personnel from AlDi Foods Inc., including production managers, supervisors, and production staff. This phase evaluates the system based on actual user experience and operational suitability.

The respondents will assess whether the system meets the organization's requirements in terms of:

- Ease of use of the interface
- Accuracy of yield and efficiency calculations
- Clarity of dashboard visualizations
- Effectiveness of stability interpretation (Stable / Not Stable)
- Usefulness of periodic reports (weekly, monthly, quarterly, yearly)
- Overall system performance in real production monitoring scenarios

Feedback gathered during UAT will be used to identify necessary improvements and refinements to ensure that the system effectively supports production monitoring, decision-making, and operational efficiency.

Through these testing procedures, the system will be validated as a reliable digital solution capable of replacing manual production monitoring methods and providing real-time analytics for improved production management at AlDi Foods Inc.

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### 3.7 Evaluation Criteria

The proposed system will undergo a formal evaluation to determine its overall effectiveness, performance, and suitability for the operational needs of AlDi Foods Inc. This assessment is designed to verify that the digital platform can reliably support the demands of a fast-paced manufacturing environment. By gathering targeted feedback from key stakeholders, the evaluation aims to prove that the system is ready for full-scale implementation.

To ensure a rigorous and standardized assessment, the evaluation will be conducted using a Likert scale questionnaire carefully adapted from the ISO 25010 Software Quality Model. Utilizing this internationally recognized standard ensures that the system is not just judged on opinion, but systematically measured against proven software quality characteristics like usability, reliability, and functional suitability. This framework provides a benchmarked, professional foundation for the entire feedback process.

A primary focus of this evaluation is determining whether the new system successfully replaces the company's legacy manual production monitoring processes. Moving away from traditional paper-and-pen logging or siloed spreadsheets represents a major shift for the production floor. The survey will closely examine how well the technology digitizes day-to-day operations and whether it creates a more streamlined, error-free workflow for the staff.

Ultimately, the evaluation will measure the system's capacity to handle advanced, data-driven manufacturing tasks. Specifically, it will assess how effectively the platform supports real-time yield analytics, extraction efficiency monitoring, and automated stability interpretation. Furthermore, it will validate the system’s ability to generate accurate periodic reporting for raw material usage, ensuring that management has the precise insights needed to optimize resources and reduce waste.

**Functionality**

Functionality refers to the degree to which the system performs its intended tasks accurately and completely. This includes the system's ability to correctly process production inputs, compute yield percentages, and generate extraction efficiency results.

The system will be evaluated based on its capability to:

- Accurately record raw material inputs and production outputs
- Compute real-time yield percentage without errors
- Calculate extraction efficiency correctly based on input-output relationships
- Automatically interpret production status as Stable or Not Stable based on defined thresholds
- Generate accurate production reports, including weekly, monthly, quarterly, and yearly summaries
- Provide real-time dashboard analytics for production monitoring

This criterion ensures that the system fulfills all required functionalities for digitized production monitoring.

**Usability**

Usability measures how easy and efficient the system is for users to operate. It evaluates the user interface design, navigation structure, and overall user experience of the system.

The system will be assessed based on:

- Ease of navigating the dashboard and system modules
- Clarity of data presentation in charts, tables, and reports
- Simplicity of entering production data such as raw materials and output yields

- Accessibility of real-time analytics and reports
- User-friendliness for production staff and managers with minimal technical background

This ensures that the system can be effectively used by AlDi Foods Inc. personnel without extensive training.

**Accuracy**

Accuracy refers to the correctness and precision of the system’s data processing and output generation. This is a critical criterion because the system is focused on production analytics and efficiency monitoring.

The system will be evaluated based on:

- Correct computation of yield percentage values
- Accurate extraction efficiency calculations
- Precise classification of production status (Stable or Not Stable)
- Consistency of reported data across different time periods
- Accuracy of raw material usage reports and historical records

This ensures that all analytical outputs reflect the actual production performance of the organization.

**Efficiency**

Efficiency measures the system’s ability to process data and deliver results with minimal delay and optimal resource usage. Since the system supports real-time monitoring, efficiency is a critical requirement.

The system will be evaluated based on:

- Speed of data processing from input to dashboard display
- Real-time updating of production metrics and analytics
- Fast generation of reports (weekly, monthly, quarterly, yearly)
- Minimal system lag during simultaneous data input and monitoring
- Effective handling of multiple production records without performance degradation

This ensures that the system supports timely decision-making in production operations.

**Reliability**

Reliability refers to the system's ability to operate consistently and correctly over time without failure. It ensures that the system remains stable during continuous use in a production environment.

The system will be evaluated based on:

- Stability of system operations during continuous data input
- Consistency of real-time analytics and dashboard updates
- Ability to recover or handle errors without data loss
- Proper storage and retrieval of historical production records
- Dependability of system outputs under normal operational conditions

This criterion ensures that the system can be trusted for daily production monitoring and long-term operational use.

**Summary of Evaluation Approach**

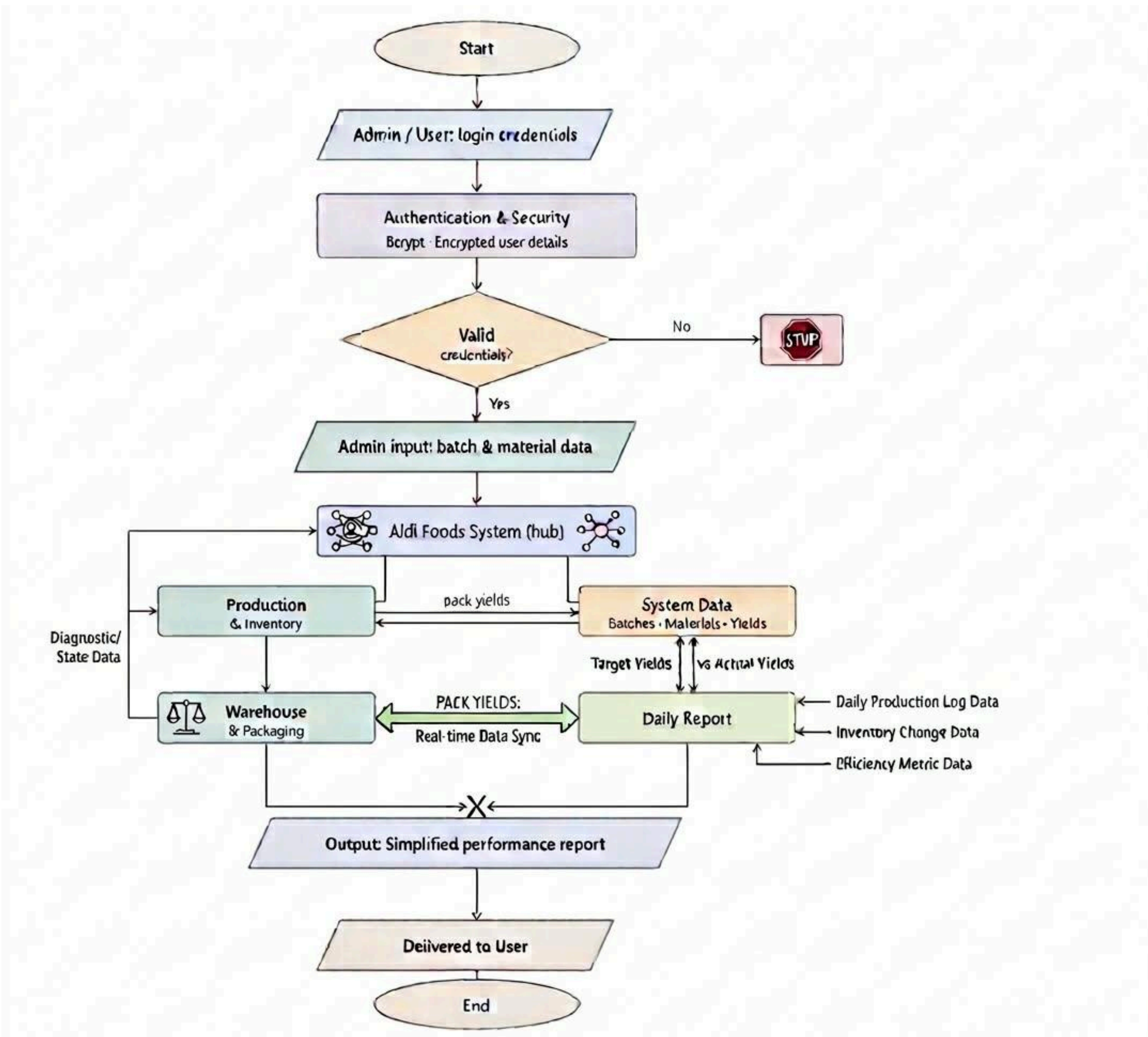
To evaluate the overall impact of the new system, selected users from AlDi Foods Inc., including frontline production staff and upper management, will complete a comprehensive Likert scale questionnaire. This structured survey will capture standardized feedback regarding user experience, system functionality, and daily operational changes. By involving both the staff on the factory floor and the decision-makers in leadership, the evaluation ensures a well-rounded perspective on how the system performs across different levels of the organization.

The data gathered from these evaluation results will serve as the primary metric to determine the overall effectiveness of the system. Specifically, the feedback will measure how successfully the digital platform replaces long-standing manual processes. Transitioning away from paper-based tracking is a major operational shift, and this assessment will pinpoint how smoothly the team has adapted to the digital transition.

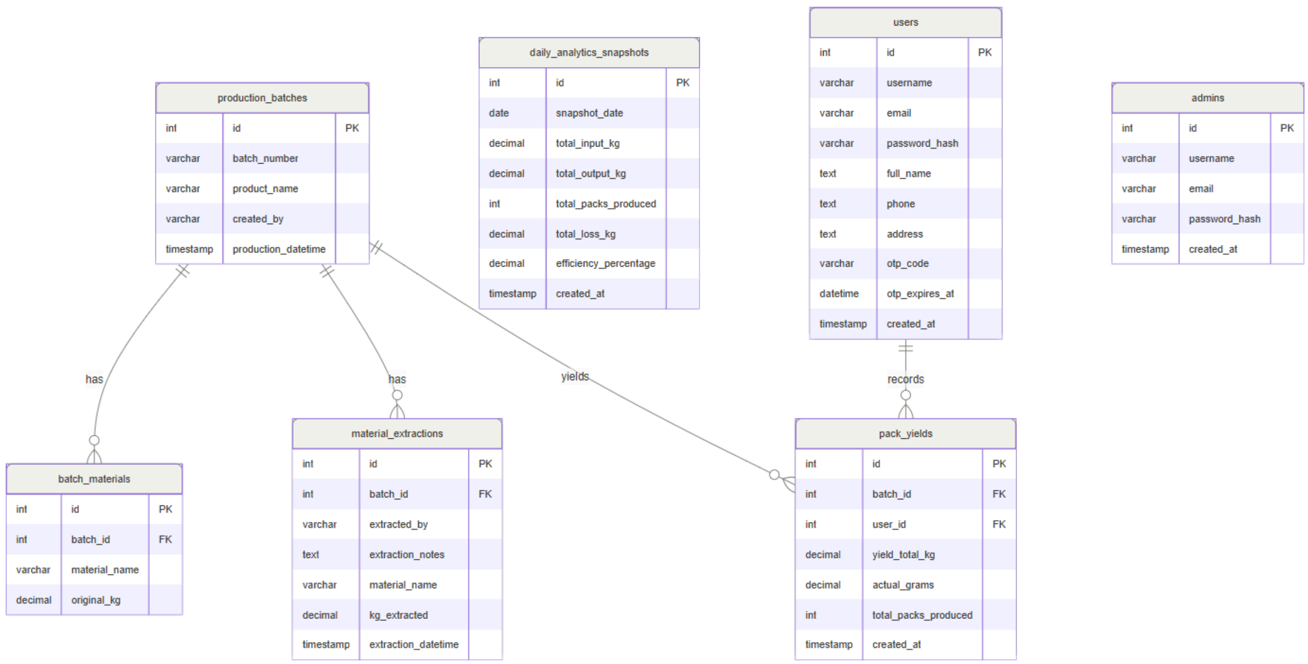
Furthermore, the evaluation will heavily focus on the system's ability to support real-time digital production monitoring. Frontline data entry and automated tracking must seamlessly translate into live updates that reflect current floor conditions. The outcome of this assessment will confirm whether the technology consistently delivers the accurate, up-to-the-minute data stream required for modern manufacturing.

Ultimately, the final results will validate whether the system successfully achieves its core operational goals. A positive evaluation will prove that the platform improves production visibility, enhances management decision-making, and drastically reduces manual data-entry errors. Most importantly, it will confirm that the system effectively supports the precise, efficient management of critical yield and extraction processes.

## Data Flow Diagram (DFD – Level 0)



**Entity Relationship Diagram (ERD) Entities:**




## - Relationships:

## 1. Users

The **Users** entity represents production personnel who interact with the system. Users are responsible for creating production batches, encoding raw material data, and recording packed yield outputs.

Users play a central role in the system because they provide the primary production data used for real-time analytics and efficiency monitoring.

**Relationships:**

- A User can create multiple Production Batches.
- A User can record multiple Pack Yield entries under different batches.

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**2. Production Batch**

The **Production Batch** entity represents a single production run or cycle within AlDi Foods Inc. Each batch contains detailed information about the production process, including raw materials used and outputs produced.

This entity serves as the main container of production data used for yield computation and extraction efficiency analysis.

**Relationships:**

- A Production Batch belongs to one User.
- A Production Batch contains multiple Batch Materials records.
- A Production Batch can generate multiple Pack Yield records.

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**3. Batch Materials**

The **Batch Materials** entity stores information about raw materials used in each production batch. This includes the quantity of raw inputs such as fruits or raw juice materials used during the extraction process.

This entity is essential for calculating production efficiency and yield percentage.

**Relationships:**

- Batch Materials belong to a specific Production Batch.
- Each Production Batch can have multiple material entries depending on production requirements.

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**4. Pack Yield**

The **Pack Yield** entity represents the final output of the production process. It records the amount of finished products (e.g., bottled or packed juice) produced from a specific production batch.

This data is crucial for real-time yield analytics and extraction efficiency computation.

**Relationships:**

- Pack Yield belongs to a Production Batch.
- A Production Batch can produce multiple Pack Yield records depending on product variations or packaging types.

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**5. Admin**

The **Admin** entity is a standalone entity responsible for system management and oversight. Admin users are not directly involved in production encoding but are responsible for monitoring system activity and managing user accounts.

**Functions of Admin:**

- Manage user accounts
- Monitor production records
- Access system reports and analytics
- Review production performance dashboards
- View stability status and efficiency results

**Relationships:**

- Admin operates independently (standalone entity) with access to all system modules.

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### 6. Analytics

The **Analytics** entity is a standalone system module responsible for processing production data into meaningful insights. It handles real-time computation and visualization of production performance.

This entity is central to the system's objective of providing digital production intelligence.

**Functions of Analytics:**

- Compute real-time yield percentage
- Analyze extraction efficiency performance
- Generate production trends and visual reports
- Classify production status (Stable / Not Stable)
- Generate periodic reports (weekly, monthly, quarterly, yearly)

**Relationships:**

- Analytics is independent but uses data from Production Batch, Batch Materials, and Pack Yield.

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## Summary of Relationships

- A **User** creates multiple **Production Batches**
- A **Production Batch** contains multiple **Batch Materials**
- A **Production Batch** produces multiple **Pack Yield records**
- **Admin** is a standalone entity for system management
- **Analytics** is a standalone module for processing and reporting production data

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## System Role in Production Monitoring

This ERD structure supports the main goal of the system, which is to digitize production monitoring at AIDi Foods Inc. It enables:

- Real-time tracking of raw material usage
- Automated yield computation
- Extraction efficiency monitoring
- Production stability classification
- Centralized reporting and analytics

By structuring the system in this way, the proposed platform ensures accurate, efficient, and real-time monitoring of production activities, replacing manual record-keeping processes and improving decision-making capabilities.

**Use Case Diagram**

The Use Case Diagram illustrates the interaction between the system and its primary actors in the proposed Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AIDi Foods Inc. It defines the functional requirements of the system and shows how each actor interacts with the system to support production monitoring, data analytics, and reporting.

The system is designed to replace manual production tracking with a centralized digital platform that enables real-time computation of yield, extraction efficiency, production stability interpretation, and automated reporting.

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**Actors**

**The system consists of two primary actors:**

● **User**

The User represents production staff or operators responsible for encoding production data into the system. Users are directly involved in recording real-time production information such as raw material usage, extraction output, and packed yield data.

The User plays a critical role in ensuring that production data is accurate, complete, and updated in real time, which is essential for generating reliable analytics and efficiency reports.

● **Administrator**

The Administrator is responsible for overseeing the entire system. This actor manages user accounts, monitors production performance, reviews analytics dashboards, and generates reports for management decision-making.

The Administrator has full access to system data and is responsible for evaluating production performance trends, monitoring efficiency levels, and ensuring data integrity across all modules.

**Use Cases:**

● **Register / Manage Admin Account**

This use case allows the Administrator to create, update, and manage admin accounts within the system. It ensures that only authorized personnel have access to sensitive production analytics and system configuration features.

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● **User Login**

Both Users and Administrators must authenticate through the login system to access their respective functionalities. This ensures system security and controlled access to production data and analytics.

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● **Production Management**

This is the core module of the system where Users input all production-related data. It supports real-time monitoring and serves as the foundation for yield analytics and extraction efficiency computation.

Sub-functions include:

- **Input Raw Materials (Kilos)**
  Users encode the quantity of raw materials used in production batches. This data is essential for computing yield and efficiency.
- **Input Extraction Kilos**
  Users record the amount of juice or extracted output obtained from the raw materials during production.
- **Input Packed Yield**
  Users enter the final packaged product output, which is used to evaluate production losses and efficiency.

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● **Inventory Yield Tracking**

This use case allows the system to automatically track and compute production yield based on raw material input and output data. It provides real-time updates on:

- Yield percentage per batch
- Production output trends
- Material utilization efficiency
- Extraction loss monitoring

This feature supports continuous monitoring of production performance and helps identify inefficiencies early.

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● **Analytics & Reporting (Administrator)**

This use case enables the Administrator to access and analyze system-generated production data. It includes real-time dashboards and structured reports that support decision-making.

Key functionalities include:

- Real-time yield analytics visualization
- Extraction efficiency performance analysis
- Production stability interpretation (Stable / Not Stable)
- Trend analysis of production performance
- Generation of periodic reports:
    - Weekly raw material usage report
    - Monthly production summary report

  - Quarterly performance analysis
  - Yearly production evaluation report

This module transforms raw production data into meaningful insights for management and operational improvement.

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● **Manage History Logs**

This use case allows the Administrator to view and manage historical production records stored in the system database. It provides access to past production batches, yield records, and efficiency reports.

This supports:

- Audit trail of production activities
- Historical comparison of yield performance
- Detection of recurring inefficiencies
- Long-term production analysis and reporting

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**System Interaction Summary**

The system operates through a structured interaction between Users and the Administrator. Users are responsible for encoding real-time production data, while the system processes this data into yield analytics, efficiency measurements, and stability classification. The Administrator then utilizes these outputs to monitor performance, generate reports, and make data-driven decisions.

Through this structure, the system successfully achieves its goal of digitizing production processes at AIDi Foods Inc., enabling real-time monitoring, automated analytics, and improved operational efficiency. -

**REFERENCES**

**2.1 Foreign Studies**

[1]**Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J. Q., Fabian, N., & Haenlein, M. (2021).** Digital transformation: A multidisciplinary reflection and research agenda. Journal of Business Research, 122, 889–901. https://doi.org/10.1016/j.jbusres.2019.09.022

**Focus:** The shift from legacy paper logs to centralized digital dashboards to eliminate "blind spots" in production

[2].**Makhmudah, S., et al. (2025).** Digitalization of production reporting in the manufacturing industry.

Focus: Transition from manual reporting to a digital system using sensors, automated data transmis/v cfsion, and dashboard visualization for real-time monitoring.

-**Mansour, M. M., & Al-Hamdani, K. S. M. (2024)**. *Key Performance Indicators for Evaluating the Efficiency of Production Processes in the Food Industry*.

Passer Journal of Basic and Applied Sciences, 6(1), 494–504.

https://doi.org/10.24271/psr.2024.450557.1555

**Focus:** Examines the use of Key Performance Indicators (KPIs) as a structured method for evaluating and improving production efficiency in the food industry. The study emphasizes the importance of distinguishing between internal and external quality indicators, particularly the use of INTppm (internal parts per million) and EXTppm (external parts per million) to measure non-conforming products in manufacturing processes. It highlights how internal production data can be used to monitor defect rates per production order, identify inefficiencies such as machine errors, tool wear, clamping issues, and process instability, and support continuous improvement in production systems.

The study further explains that KPI-based evaluation enables organizations to detect waste, improve quality control, enhance machine reliability, and strengthen operational decision-making. It also stresses the importance of integrating systematic monitoring, employee training, preventive maintenance, and process optimization strategies to reduce internal defects and improve overall production performance in food manufacturing environments.-

-**Aftab Siddique, A., Gupta, A., Sawyer, J. T., Huang, T.-S., & Morey, A. (2025)**. *Big DataAnalytics in Food Industry: A State-of-the-Art Literature Review*. npj Science of Food. https://doi.org/10.1038/s41538-025-00394-y

**Focus:** Reviews the application of artificial intelligence (AI), machine learning (ML), and big data analytics in the food industry, emphasizing their role in improving production efficiency, quality control, food safety, supply chain management, and sustainability. The study highlights how real-time data collection, predictive analytics, neural networks, and computer vision systems are used to optimize food processing operations, support decision-making, and reduce waste, while also discussing challenges such as data quality, system integration, and cybersecurity in digital food manufacturing systems.

**-Rivadeneira, J. P., Wu, T., Ybañez, Q., Dorado, A. A., Migo, V. P., Nayve, F. R. P., & Castillo-Israel, K. A. T. (2023).** Ultrasound-assisted extraction of pectin from *saba* banana peel waste: characterization, emulsifying properties, and rheological behavior. *Food Research.*

**Focus:** This study evaluates ultrasound-extracted pectin from *saba* banana peel waste as a functional emulsifier compared to commercial pectin. Results showed that UEP had lower methoxyl but higher protein content, leading to better emulsion stability under acidic conditions, higher oil fractions, and increased pectin concentration. It also formed smaller droplets, showed higher viscosity, and reduced lipid digestion.

Overall, the study concludes that banana peel-derived pectin is a sustainable and effective alternative for food emulsion applications.

**2.2 Local Studies**

**[3]Bagaforo, H., & Albason, A. (2024).** Digital transformation journeys of MSMEs

in Iloilo Province: Enablers and barriers. 19th European Conference on Innovation and Entrepreneurship (ECIE 2024).

**Focus:** Evaluates the readiness of local businesses in Iloilo to adopt new technology and the barriers (like technical literacy) they face.

**[4]De Leon, C., Demos, E., Manzanares, M. S., Sanorjo, T. A., & Villamor, J. (2017).** Design and development of an online ordering system for Iloilo Supermart (Unpublished bachelor's thesis). Central Philippine University, Iloilo City. https://repository.cpu.edu.ph/handle/20.500.12852/2281

**Focus:** A technical blueprint for secure data management and real-time monitoring within a local commercial context (Iloilo Supermart).

**-Palero, H. C. F. P., B. H. M. I., R. P. S. B., M. C. F., & E. J. R. C. (2026)**. *Green extraction and sustainability assessment of carotenoids from carrot peels using soybean oil*. Davao Research Journal.

**Focus:** This study optimizes carotenoid extraction from carrot peels using soybean oil as a green solvent. It applies factorial design and statistical modeling to analyze the effects of liquid-to-solid ratio and temperature, identifying the liquid-to-solid ratio as the most significant factor. Optimal conditions yielded up to 84.22 ± 5.56 μg β-carotene/g.

A sustainability assessment using Path2Green showed a moderate green score (0.593), with strengths in solvent safety and biomass utilization but limitations in energy use. The study concludes that soybean oil is a viable eco-friendly solvent for carotenoid extraction.-

https://davaoresearchjournal.ph/index.php/main/article/view/575/306

-**Manalo, J. A. (2021).** Digital Agriculture – How Can We Make It Work? Rice Science for Decision-Makers, 9(June 2021). Department of Agriculture–Philippine Rice Research Institute (DA-PhilRice).

**Focus:** Examines the implementation of Digital Agriculture in the Philippines through the use of Information and Communications Technologies (ICTs) to modernize agricultural production and improve farmers' access to information. The study highlights the gap between ICT access and actual technology utilization among Filipino farmers, identifying barriers such as ICT anxiety, low digital literacy, and the aging farming population. It discusses the DA-PhilRice Infomediary Campaign, which mobilized high school students to assist farmers in accessing agricultural information through digital platforms.

The study demonstrates how ICT-based tools, mobile applications, digital information systems, and knowledge-sharing initiatives can enhance information dissemination, support data-driven decision-making, encourage technology adoption, and improve operational efficiency in agricultural activities while promoting inclusive digital transformation in rural communities.-

**-Barroga, K. D., Rola, A. C., Depositario, D. P. T., Digal, L. N., & Pabuayon, I. M. (2019).** *Determinants of the Extent of Technological Innovation Adoption Among Micro, Small, and Medium Food Processing Enterprises in Davao Region, Philippines*.
Philippine Journal of Science, 148(4), 825–839.

**Focus:** This study examines the factors affecting the extent of technological innovation adoption among MSMEs in the food processing industry in Davao Region, Philippines. It analyzes how personal and organizational characteristics influence adoption of DOST-SETUP innovations such as food safety training, GMP assessment, and equipment-related upgrades, and identifies key determinants including sex of owner, educational attainment, business scale, and type of market.-

**Ang, R. C. A., Amongo, R. M. C., Paras Jr., F. O., Suministrado, D. C., & Peralta, E. K. (2022)**. *Development of banana peduncle juice extractor for ethanol and fiber production*. Philippine Journal of Agricultural and Biosystems Engineering, Vol. 18, No. 1 (June 2022).

**Focus:** Designs and evaluates a specialized crushing-roll juice extraction machine for banana peduncles aimed at improving bioethanol production and fiber recovery from agricultural waste. It examines how roll speed and crushing load affect machine efficiency, juice yield, fiber recovery, and fuel consumption, and identifies optimal operating conditions using response surface methodology.

**-Jiang, M., & Po, S. M. (2023).** *The effect of human resource practices on employee retention in the food manufacturing industry in the Philippines: The moderating role of the work environment. International Journal of Research Studies in Management.*

**Focus:** This study investigates the effect of human resource practices on employee retention in the food manufacturing industry in the Philippines and examines the moderating role of the work environment. The human resource practices analyzed include training and development, employee participation in decision-making, compensation, job security, and performance appraisal.-

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### 2.3 Related Literature

These sources provide the technical and institutional framework for your project's implementation.

**Salamat, R., & Mendoza, J. (2025).** Implementation of advanced food technology and centralized management at the USWAG Iloilo City Nutrition Center. Department of Science and Technology Region VI.

**Focus:** A local case study on digitizing food production workflows to maintain high-level oversight and food safety standards.

**[5]Somsen, J., Capelle, A., & Tramper, J. (2004).** Production Yield Analysis (PYA) in food processing systems.

**Focus:** Development of a structured method to evaluate raw material efficiency by distinguishing necessary and avoidable losses using a Yield Index to identify production inefficiencies and hidden waste.

[6]**Charbonnier, S., Béranger, J., & Cognon, R. (2005). T**rends extraction and analysis for complex system monitoring and decision support in industrial and medical environments. Engineering Applications of Artificial Intelligence.

**Focus:** A foreign study on real-time trend extraction techniques for monitoring complex industrial and medical systems by converting raw data into simple symbolic trends (increasing, decreasing, steady) to improve decision-making, reduce operator workload, and support real-time process monitoring in manufacturing environments.

[7] **Davies, P. L., Fried, R., & Gather, U. (2003).** Robust signal extraction for on-line monitoring data. Computational Statistics & Data Analysis.

**Focus**: A foreign study on robust statistical methods for real-time signal extraction in online monitoring systems, focusing on separating meaningful trends from noise, outliers, and measurement errors in time-series data. The study evaluates regression-based techniques such as L1 regression, repeated median, and least median of squares to improve accuracy in detecting trends, level shifts, and changes in system behavior for critical applications like healthcare monitoring and industrial process control.

**[8]Saflor, C. S. R., & Noroña, M. I. (2021).** Total Productive Maintenance in the Philippine Rice Production Sector. Proceedings of the International Conference on Industrial Engineering and Operations Management, São Paulo, Brazil.

**Focus:** A study that evaluated maintenance practices in the Philippine rice production sector and proposed a Total Productive Maintenance (TPM)-based framework to improve machine reliability, reduce operational and maintenance costs, minimize equipment breakdowns, and enhance overall productivity through preventive maintenance, worker training, and systematic equipment management.

[9]**Desai, S. S. (2020)**. Design, Development and Performance Evaluation of Juice Extractor. Master of Technology Thesis, Mahatma Phule Krishi Vidyapeeth, Rahuri.

**Focus:** Engineering design and performance evaluation of a hydraulic juice extractor system for pomegranate arils and bottle gourd, analyzing how size reduction, applied pressure, and filter bag pore size affect juice yield, extraction efficiency, and extraction losses to optimize juice processing performance.

**Note to the instructors handling this subject**

## CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.